Chesterfield County Land Development
Phase II Transformation Support

Records Management Final Report

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1. Executive Summary

In addition to complying with Library of Virginia (LVA) requirements, Chesterfield County operates a records management function to:

- Preserve institutional memory
- Organize and manage information and knowledge
- Facilitate accountability and transparency
- Provide evidence of government activities
- Enable and support investigations, litigations, audits, and fulfillment of Virginia Freedom of Information Act (VFOIA) requests

IT Cadre was tasked to assess the County's records management activities to:

- Provide an end-to-end view ("Current State Visualization") of Chesterfield County Records Management operations, governance, processes, systems, task owners, and Stakeholders (e.g., identifying, classifying, storing, securing, retrieving, tracking and destroying or permanently preserving records)
- Support the development and prioritization of recommendations to improve the efficiency and effectiveness of Chesterfield County Records Management activities
- Provide a foundation for developing an actionable records digitization and materials handling improvement plan

The Current State Visualization process began with developing a 15-day draft of the visualization based on research and document reviews. The draft content was iteratively updated, corrected, expanded, improved, and enhanced through a cycle of interviewing County records management stakeholders and updating the visualization based on each interview. This iterative process led to broad, shared agreement that the visualization represents "ground truth" with respect to the current state of records management in the County. It also led to the development and validation of key reads and observations related to the current state.

Once the current state of records management was understood, IT Cadre worked with County Leadership to define strategic goals and outcomes for the future (see section 4)… the desired state of records management in the County that could be realistically achieved in the next two years (by the end of 2026).

Based on observations made during the research, document reviews, and interviews (see section 2.3), and consideration of the strategic goals and outcomes, IT Cadre developed the 19 recommendations shown below (see detailed descriptions in section 5), organized by implementation priority.

Priority	Recommendation
1	Enhance Records Management Officer Roles & Responsibilities
	Investigate Perceived Need for Duplicate Records
	Enhance County-wide Electronic Records Mgmt Policy
	Conduct Audits and Provide Support
	Develop Records Locator Inventory (LVA RM-20 Form)
2	Exchange More Electronic Records, Fewer Hardcopy
	Define Records Mgmt Plans for Software Platforms
	Pursue Automated Retention Schedule Assignment and Records Destruction
	Provide Training
	Investigate Wet Signature Record Requirements
	Manage LVA Records Retention Schedule Deviations
	Define and Track Measures and Metrics
	Review, Digitize, and Destroy Hardcopy Records
	Control Off-Site Storage Costs
	Develop Enhanced Records Transfer Process and Instructions
	Define Electronic Record Format Requirements
	Add Box# to LVA Records Destruction Workflow
3	Relocate Records Management Officer Authority
	Conduct Awareness Campaign

2. Current State Visualization Overview

2.1 Project Purpose

The purpose of developing the Current State visualization for Records Management was to provide an end-to-end view of Chesterfield County Records Management operations, governance, processes, systems, task owners, and stakeholders (e.g., identifying, classifying, storing, securing, retrieving, tracking and destroying or permanently preserving records).

Based on this view, a prioritized list of recommendations to improve the efficiency and effectiveness of Chesterfield County Records Management activities appears in section 6.

2.2 Visualization Development Process

The Current State Visualization process began with developing a 15-day draft of the visualization based on research and document reviews. The draft content was iteratively updated, corrected, expanded, improved, and enhanced through a cycle of interviewing County records management stakeholders and updating the visualization based on each interview. This iterative process led to broad, shared agreement that the visualization represents "ground truth" with respect to the current state of records management in the County.

Appendix A lists Chesterfield County employees (Record Owners and Record Administrators) who were interviewed as part of the iterative review and validation of the Current State Visualization.

Appendix B shows some example questions that were asked during the initial interview with each Department / office, typically asked of the Record Administrator.

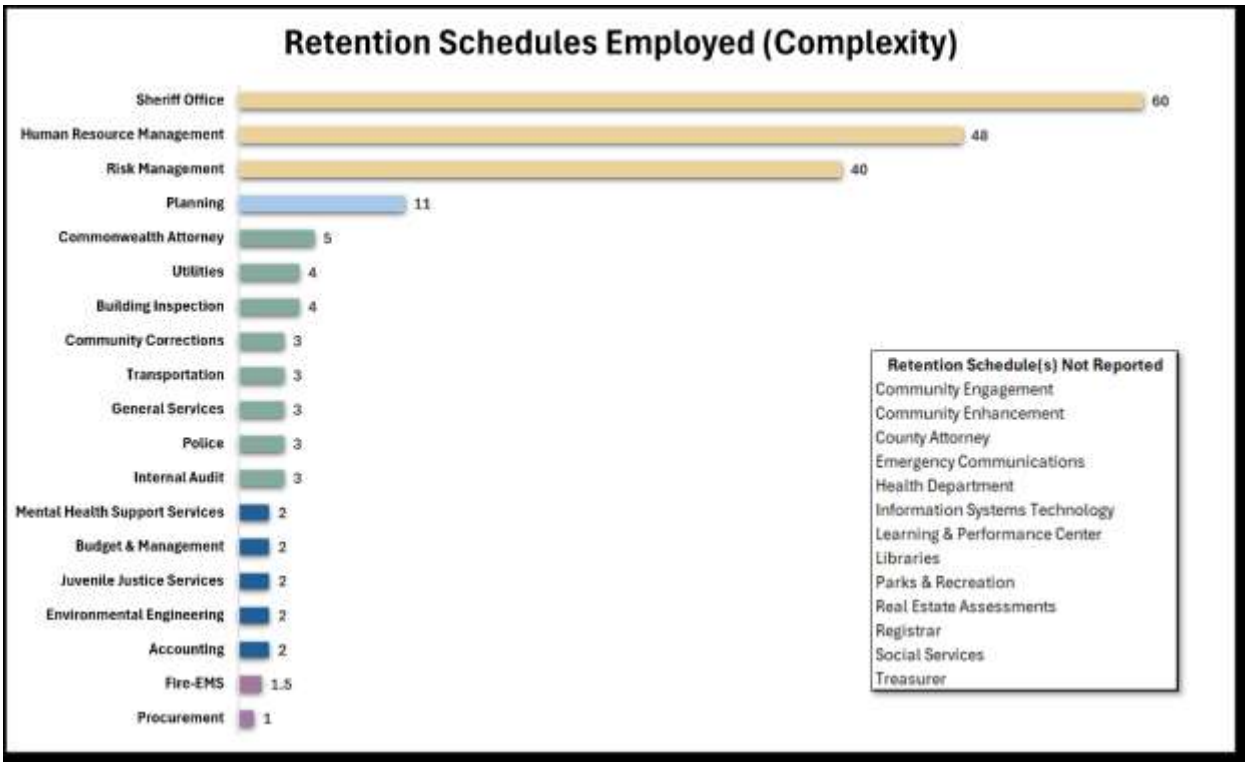
Based on observations made during the research, document reviews, and interviews (see section 2.3), and consideration of the strategic goals and outcomes, IT Cadre developed the 19 recommendations shown below (see detailed descriptions in section 5), organized by implementation priority.

2.3 Graphs & Data

As part of the data collection and interview process, records management-related data was collected. The graphs in this section provide some insight into the County's records management processes and inform the Key Reads & Observations and the Recommendations.

2.3.1 Retention Schedules (Complexity)

The graph, below, shows the quantity of retention schedules (combinations of schedule and record series) used by each department / office. Using a greater quantity of schedules introduces more complexity into the records identification, tracking, and storage process. However, using more schedules is also aligned with a more mature and rigorous records management process.

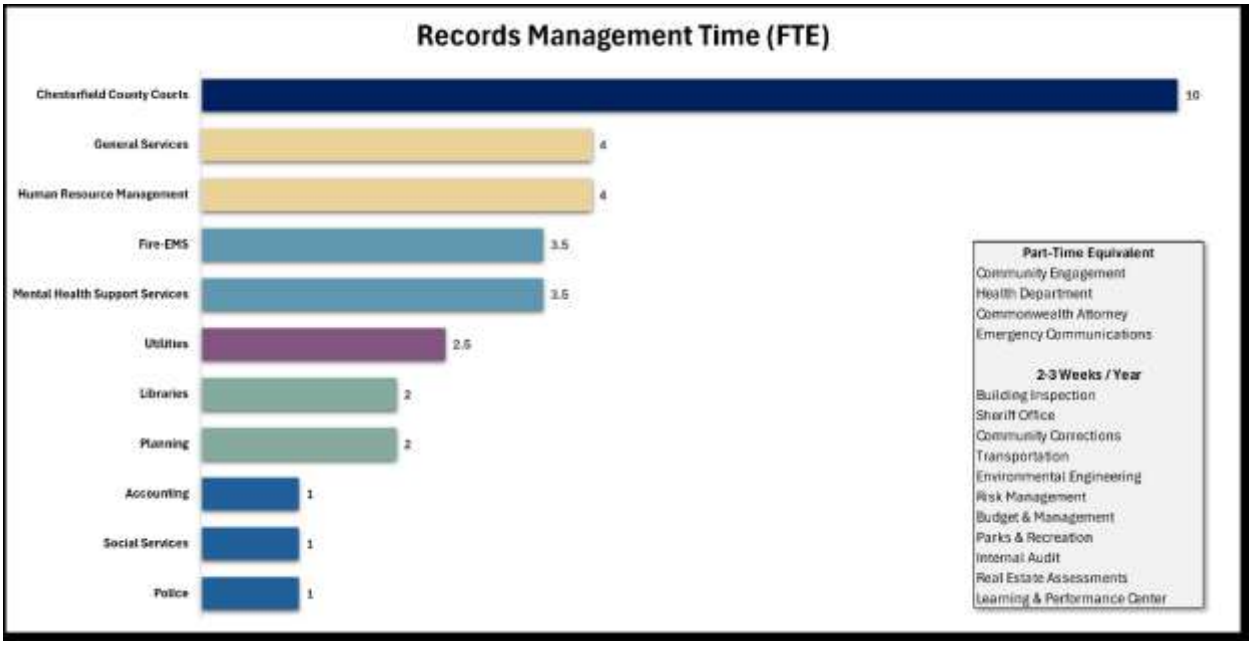


2.3.2 Record Management Time

The graph, below, shows an estimate of the amount of time spent by each department / office on records management. Estimates were provided by Record Owners and Record Administrators from each department / office. This estimate includes the time spent performing the following activities:

- Scanning records
- Requesting boxes
- Requesting bar code labels for boxes
- Organizing records in boxes
- Completing and submitting records transfer forms
- Submitting requests to retrieve boxes from off-site storage

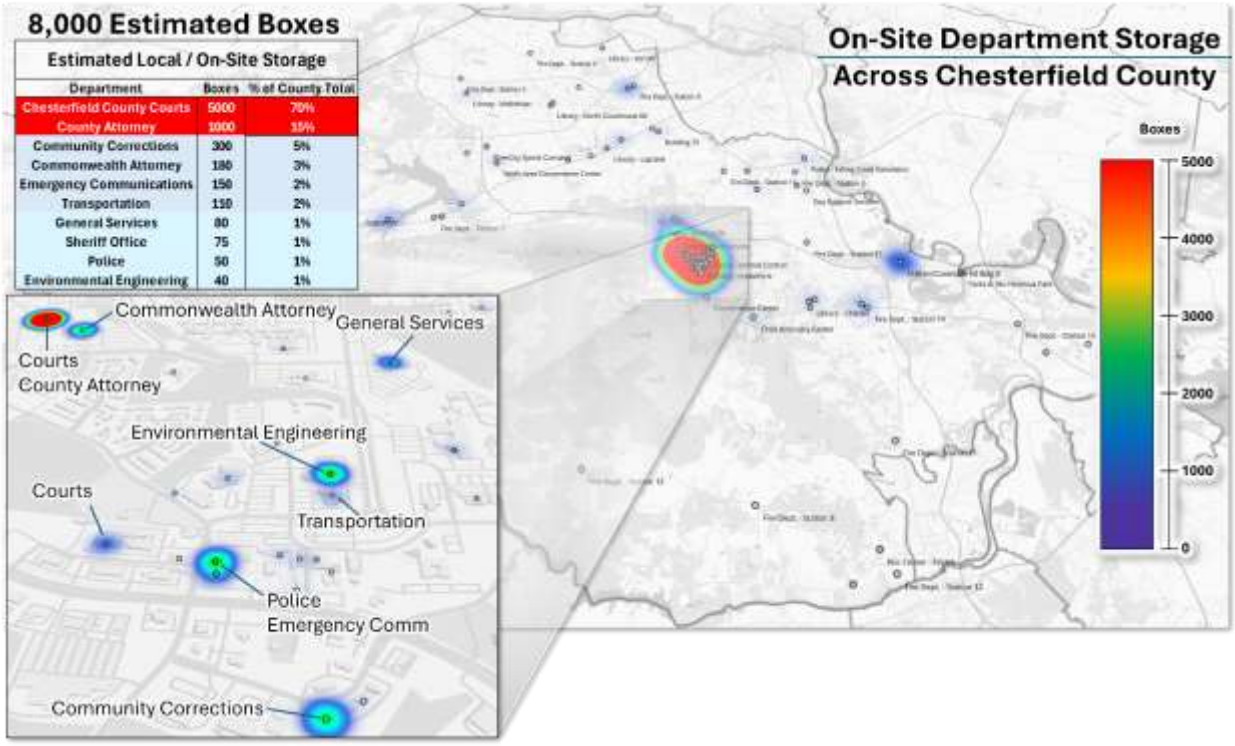
Organizations with more than 1 FTE allocated to records management are scanning records. Improvement efforts related to records digitization should include the departments / office that do the most scanning, as shown here.



2.3.3 Current On-Site (Local) Storage

The graphic, below, shows the locations across Chesterfield County where records are stored. The quantity of records stored at each location appears as a color-coded heat map.

This map illustrates the complexity of records management in Chesterfield County. There are many organizations storing lots of records in many different locations. Improvement efforts related to reducing locally stored records should include the departments / offices with the most locally stored records, as shown here.



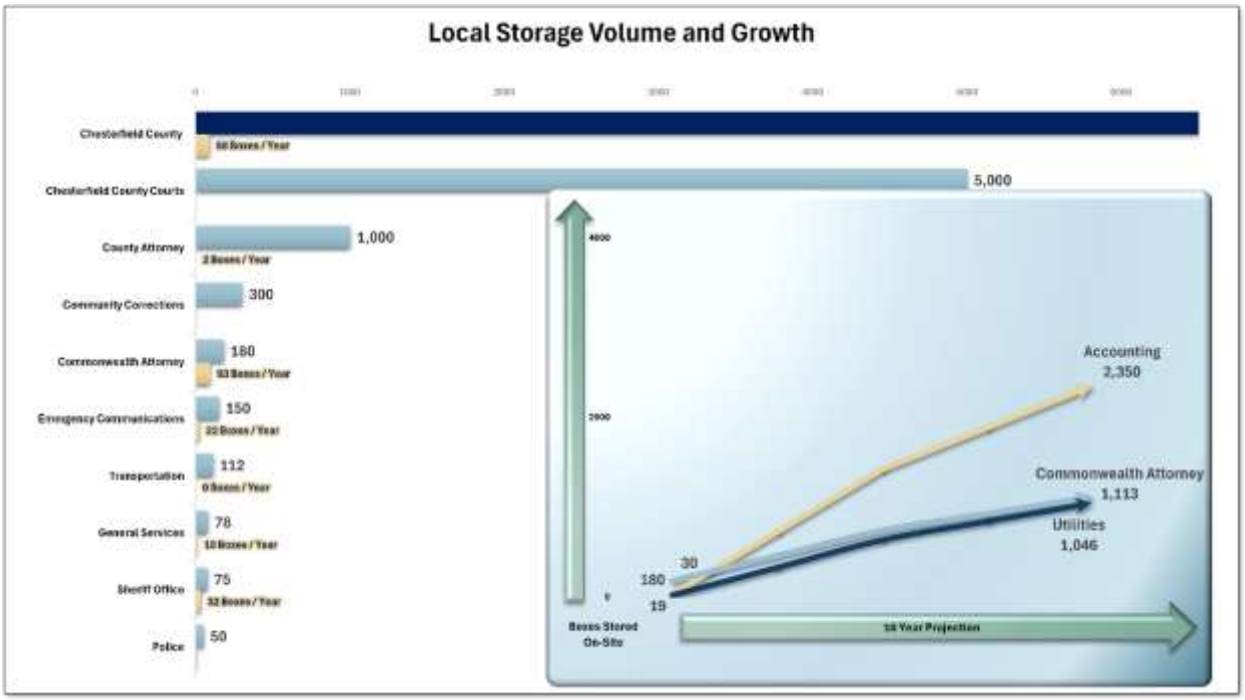
2.3.4 On-Site (Local) Storage Growth

On the left side of the graphic, below, shows the current volume and rate of growth in the quantity of locally stored boxes of records, by department / office. The smaller graphic on the right side shows a projection of the quantity of locally stored boxes at the end of 2034, 10 years from now, for the departments / offices whose local storage is growing the fastest.

The blue bars show the current quantity of locally stored records, by department / office.

The tan bars show the growth in locally stored boxes of records per year. The annual growth in boxes of locally stored records is calculated based on the quantity of locally stored boxes that are destroyed or sent to VRC each year, subtracted from the quantity of new boxes of records being stored locally.

Departments / offices whose local inventories are growing will have to allocate more space for local storage or increase the quantity of boxes they send to VRC.

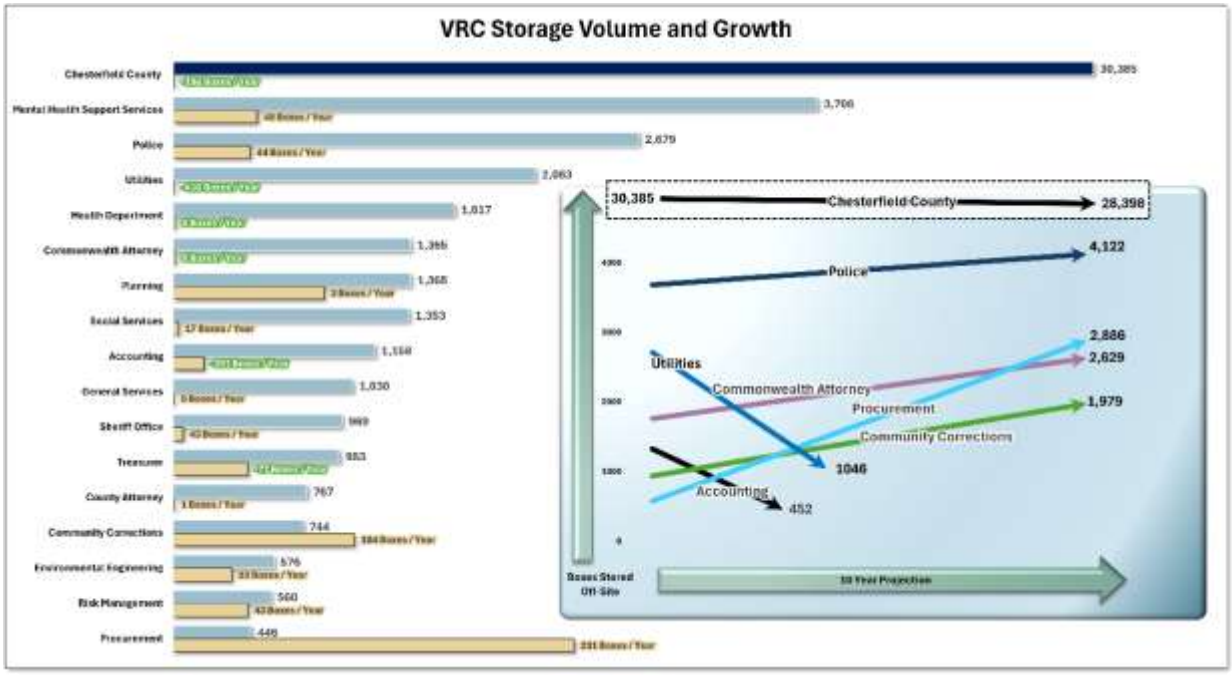


2.3.5 VRC Storage Growth

On the left side of the graphic, below, shows the current volume and rate of growth in the quantity of boxes of records stored at VRC, by department / office. The smaller graphic on the right side shows a projection of the quantity of boxes at the end of 2034, 10 years from now, for the departments / offices whose VRC is growing the fastest.

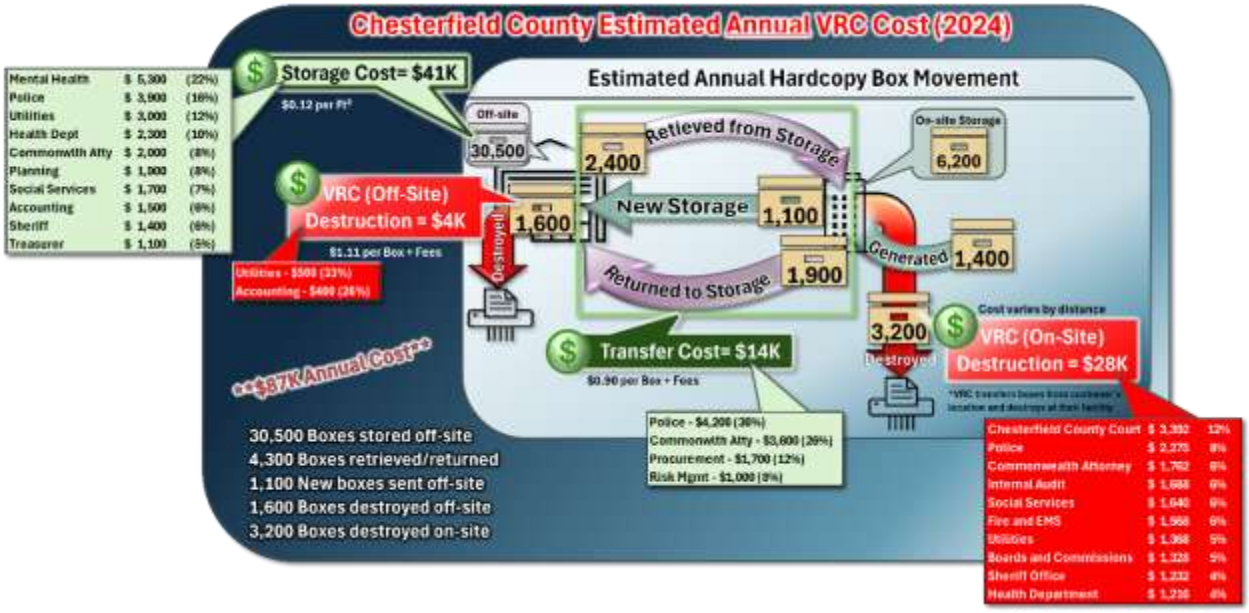
The blue bars show the current quantity of records currently stored at VRC, by department / office.

The tan bars show the growth in boxes of records at VRC, per year. The annual growth in boxes is calculated based on the quantity of boxes that are destroyed at VRC each year (those that reached the end of their retention period), subtracted from the quantity of new boxes of records being sent to VRC each year. Note that the bar graph line labels are color coded to show volume increases in tan and volume decreases in green.



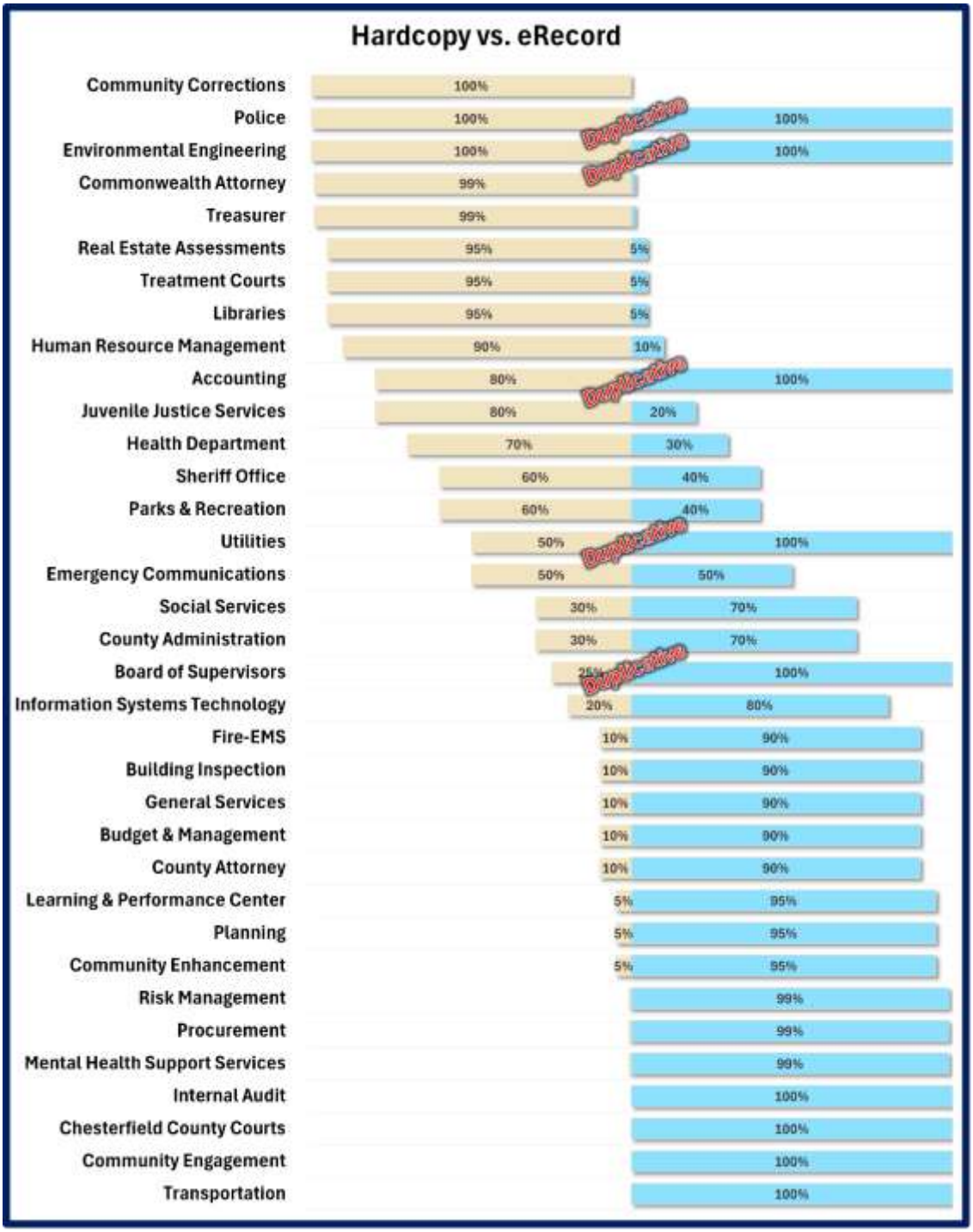
2.3.6 Off-Site Storage Cost Growth

The graphic, below, shows a breakdown of the current costs for off-site storage of records at VRC, based on the most current data. Only 9 months of data in 2024 was available. This data was extrapolated to estimate annual costs. The annual cost to store County records at VRC is estimated to be $87K.



2.3.7 Hardcopy vs. Electronic Records Generation

The graphic, below, shows the mix of hardcopy (the bar on the left) and electronic (the bar on the right) records generated by department. In some cases, a department will digitize a hardcopy record and then keep both the hardcopy and electronic copies as records. This appears on the graph as a mix of hardcopy and electronic records that total more than 100%.



3. Key Reads & Observations

These key reads and observations were developed based on research and interviews conducted with Record Owners and Record Administrators from most of the County departments and offices. These key reads and observations will inform the Recommendations that appear later in this document.

The key reads and observations are organized into 4 groups:

- Records Identification
- Records Storage & Retrieval
- Records Retention & Destruction
- Records Digitization

3.1 Records Identification

3.1.1 Maintaining Multiple Copies of Records

- Some departments / offices identify and manage multiple copies of a single work product as records.
    - Accounting, Building Inspections, Deputy Administration, Environmental Engineering, Human Resources, Police (including Animal Control), Procurement, Sheriff, and Utilities identify and manage multiple copies of a single work product as records. Often this is in the form of managing a hardcopy record and a duplicate electronic record. Because their RMS can only hold a limited number of records for each case, they often have to scan the case records into LaserFiche and then then rescan a subset of those records into RMS.
    - Some interviewees report that they and/or their organization's employees would benefit from additional training to help them better understand how to identify a record and their responsibilities associated with managing their records.
    - When multiple organizations have interest in a record, identification of the record owner can be unclear, resulting in duplicate "official" copies of the same record.
    - Some departments / offices keep copies of HR-related records, duplicative of what HR keeps.

3.1.2 Lack of Separation for Records vs. Non-Records

- Some departments / offices place records into electronic storage unsorted and comingled with other work products, making those records difficult to identify.
    - Accounting, Community Engagement & Resources, Health, Social Services, and General Services Waste and Resource Recovery do this.

3.2 Records Storage & Retrieval

3.2.1 Records Retained Beyond Retention Period

- Some departments / offices retain records for longer than their Library of Virginia (LVA) retention schedules specify due to: (1) conflicting retention requirements from other

agencies, (2) the desire to keep them for historical reference, (3) retention schedule has inappropriate retention period, or (4) uncertainty in the rules associated with their destruction.

- Most departments / offices report that they do not destroy electronic records in accordance with an assigned retention schedule… or at all in some cases.
- Community Enhancement keeps case files for reference, beyond the prescribed retention period.
- In some cases, records are retained beyond the LVA retention schedule do to other applicable retention requirements
  - Misc: State or Federal government retention requirements
  - Accounting: IRS and SEC retention requirements
  - Police-Animal Control: Virginia Board of Veterinary Medicine retention requirements
  - Mental Health: CSB local series and Va DSS Mental Health series
- In some instances, a case file or other logical grouping of records includes records with different retention periods (e.g., Juvenile Justice Services, Police, Treasurer). Typically, the entire case file is kept together in a single box with the longest retention period assigned to the box, resulting in keeping some records beyond their retention period.
- Affected departments / offices include:
  - Accounting
  - Budget & Management
  - Building Inspection
  - Community Engagement
  - Community Enhancement
  - Environmental Engineering
  - Learning & Performance Center
  - Fire-EMS
  - General Services
  - Juvenile Justice
  - Mental Health Support Services
  - Police
  - Police – Animal Control
  - Real Estate Assessments
  - Sheriff
  - Social Services
  - Transportation
  - Utilities

3.2.2 Successful Records Digitization & Destruction

- Department / office-wide initiatives to digitize and destroy hardcopy records appear to have been highly successful in reducing hardcopy storage volume.
  - Departments / offices that have done a digitize and destroy effort include Budget and Management, Building Inspections, Community Engagement, Learning & Performance, Procurement, Risk Management, Transportation, and Real Estate Assessments.
  - Transportation is in the process of retrieving boxes from VRC, digitizing records to be kept, and destroying records whose retention period has expired and destroying documents that aren't really records that were included in the box.
  - In some cases, this digitize and destroy effort is limited to locally stored records, while others retrieved all of their records from VRC and digitized & destroyed as part of this effort.

- When Planning retrieves a box from VRC, they digitize and destroy the records in the box. This appears to be a good model.

3.2.3 Hardcopy Retention Required for Signed Records

- Several departments / offices reported a requirement to keep signed documents as hardcopy records, with no option to digitize and destroy the hardcopy.
  - Community Enhancement, Learning & Performance, Parks & Recreation, Planning, Procurement, Real Estate Assessments keep signed documents as hardcopy, citing concerns about the legality / appropriateness of digitizing them and destroying the hardcopy.

3.2.4 Confusing Records Transfer Process

- Several departments / offices reported that the records transfer process is cumbersome and confusing.
  - Departments / offices reporting this include Accounting, Budget & Management, Fire-EMS, Mental Health Support Services, and Risk Management.

3.2.5 Lost Records

- Some departments / offices report issues retrieving records from VRC – one department reported 9 missing boxes of records.
  - This issue is reported by Board of Supervisors, Emergency Communications, Police, and Procurement.
  - Nine missing VRC boxes were discovered because Procurement retrieved all of their boxes from VRC. Other departments / offices may have missing boxes, but there are not known because they have never requested retrieval.
  - Several departments / offices reported occasional delays in box retrieval due to the records being missing temporarily, but all were eventually found.

3.3 Records Retention & Destruction

3.3.1 Inability to Confirm Correct Records Destruction

- The LVA workflow task that is sent Record Owners, which requests their approval to destroy records, does not specify the box number or otherwise describe the records to be destroyed. They must approve the workflow task to move forward, but without any positive confirmation of *which* records will be destroyed.

3.3.2 Lack of Retention Schedule Assignment

- Most departments / offices do not assign retention schedules to electronically-stored records.
  - Many indicate that they keep electronic records forever, but have not mapped the various kinds of records to a specific LVA retention schedule (e.g., electronic training records in Fire-EMS)
  - The Learning and Performance Center keeps some electronic records as Permanent for “historical reference”, despite shorter retention periods prescribed by LVA.

- Fleet Management does not assign retention schedules to hardcopy records.
- Community Engagement & Resources keeps all work products permanently. The do not breakout records from non-records – or one could argue that they view all work products as records.
- Most County electronic records are not assigned a retention schedule.
- Capital Projects & Programs keep records permanently, but do not assign a retention schedule.

3.3.3 Lack of eRecord Retention Period Tracking

- Most departments / offices have difficulty in efficiently tracking retention periods for electronic records.
  - Most departments / offices are not tracking the retention period of electronic records, so they are unaware when retention periods expire.
  - Commonwealth Attorney uses hardcopy records exclusively because it is difficult to assign retention schedules to electronic records and also difficult to delete records in the system(s) in a semi-automated way. It is faster and easier for them do deal in 100% hardcopy as a result.

3.3.4 Lack of eRecord Destruction

- Most departments / offices do not destroy electronic records due to uncertain rules and the inherent difficulty with efficiently destroying either individual or multiple files.
  - Most County electronic records are not destroyed in accordance with an assigned retention schedule
  - Commonwealth Attorney uses hardcopy records exclusively because it was difficult to assign retention schedules to electronic records and deleting records in the system(s) was also difficult. It is faster and easier for them do deal in 100% hardcopy as a result.
  - While most departments / offices can find and destroy the "official" copy of a hardcopy record when its retention period expires, they cannot do the same for "unofficial" copies of records that are stored locally by employees and kept beyond the prescribed retention schedule.
  - Planning is seeking an automated way to destroy expired records from LaserFiche, without having to destroy one record at a time.

3.3.5 No RM-3 Form for Local Records Destruction

- Some departments / offices destroy locally-stored hardcopy records without completing the RM-3 process.
  - The Commonwealth Attorney, Community Enhancement, Emergency Communications, Internal Audit, and Risk Management store hardcopy records with relatively shorter retention schedules locally. They destroy these records when the retention period is complete. However, they do not complete an RM-3 form to document its destruction. Real Estate Assessments has historically not completed an RM-3 form when destroying local records, but are starting to now.

- Internal Audit wasn't aware of any policies or procedures that speak to destroying locally-stored records (hardcopy or electronic)

3.3.6 Lack of Authority and Resources

The County Records Management Officer, as a subordinate organization within the IST, lacks sufficient authority and resources needed to make and enforce strategic Records Management-related decisions.

- The Records Management Officer lacks sufficient authority to reject requests for records management deviations received from divisions / offices (e.g., records retention period extensions)
- The Records Management Officer does not have sufficient resources to provide department- / office-specific direction, require and provide training, and perform other activities required to manage a robust records management program.
- The Records Management Officer lacks the necessary authority to enforce compliance with records management requirements on the County departments.

3.4 Records Digitization

3.4.1 Lack of Records Management in Software Platforms

- Most software platforms lack an effective records management plan for identifying and handling stored records – no capability to properly tag metadata to individual or groups of records, which prevents an efficient destruction process.

3.4.2 eRecord File Format Confusion

- Record Administrators report confusion as to the rules associated with required file format standardization for storing electronic records (e.g., PDF vs TIFF).
    - Budget & Management, Environmental Engineering, and Human Resources need guidance in selecting an acceptable electronic record format (e.g., PDF vs TIFF) that meets records management requirements and is supported by existing software systems.

3.4.3 Police RMS-LaserFiche Records Duplication

- The Police department's Records Management System (RMS) restricts the number of scanned documents combined and attached into an official electronic record, forcing the department employees to scan hardcopy twice – once into LaserFiche and again into RMS.
- The Police department's Records Management System (RMS) is often unable to store the complete set of files associated with an electronic record, forcing the department employees to scan hardcopy twice – once into LaserFiche and again into RMS.

3.4.4 Digitization Prioritization

- Chesterfield County lacks clearly defined criteria for selecting and prioritizing records to be digitized.

- There is inconsistency across departments / offices as to which records are digitized
- The county lacks a cost analysis to be used to inform records digitization criteria and planning
- Many departments / offices would like to digitize more of their current and historical records, but lack the time and resources to do so.

4. Strategic Goals & Outcomes

These strategic goals and outcomes were developed to envision what success in records management would look like for the County, two years from now (by the end of 2026). These goals and outcomes will inform the Recommendations that appear later in this document.

The strategic goals and outcomes are organized into 4 groups:

- Compliance & Leadership
- Record Formats
- Cost
- Integration & Automation

4.1 Compliance & Leadership

Records management compliance and leadership is led by individuals serving in several key roles, such as Records Management Officer, Records Owner, Records Liaison. See Appendix F for a brief overview of these roles.

4.1.1 LVA Records Management Requirements Compliance

Chesterfield County is fully compliant with all LVA records management requirements.

4.1.2 Records Management Leadership Assigned to the Right County Department

Records Management Leadership is assigned at an organization level with sufficient authority and resources to ensure ongoing improvements in records management efficiency and effectiveness.

4.1.3 Electronic Records Management Rules Defined & Followed

Departments / offices understand and follow clearly defined rules for how to store, assign a retention schedule, and destroy electronic records.

4.1.4 Retention Exceptions Managed

All requested record retention period extensions are documented in a list, owned by the Records Officer, and adjudicated by the CIO and the DCAs, collectively.

4.1.5 Measurement Data

Complete and accurate measurement data is available to provide insight into records management operations.

4.2 Record Formats

4.2.1 Overall Trend Toward Electronic Records

Departments / offices are improving their ratio of hardcopy to electronic records.

4.2.2 Records Stored as Hardcopy or Digital not Both (with limited exceptions)

For new & existing records, departments / offices no longer keep both hardcopy & digital versions as records.

4.3 Cost

4.3.1 Rate of VRC Cost Growth Reduced

The total amount paid to VRC is trending lower.

4.4 Integration & Automation

4.4.1 Software Platform Integration for Records Management

Software Platforms that contain records are integrated and interoperable (e.g., Police RMS is integrated with LaserFiche so that records scanned into one system are accessible from the other system).

4.4.2 Automated Records Retention Schedule Assignment and Record Destruction

The County leverages automated or semi-automated mechanisms to assign retention schedules to electronic records and to destroy expired electronic records. This includes investing in new SW platform capabilities (i.e., the proposed Accela update for enhanced records management) and leveraging capabilities like automated file tagging in SharePoint.

4.4.3 Legal Hold Tagging

The County has the ability to tag electronic records that are on legal hold.

5. Recommendations

The recommendations that appear in this section were developed in response to Key Reads and Observations and the Strategic Goals & Outcomes that appear in this document. A mapping of key reads & observations to recommendations appears in Appendix E.

The recommendations are organized into 4 groups:

- Enhance Records Management Oversight and Leadership
- Reduce the Quantity of Hardcopy Records
- Compliance with LVA Records Retention Requirements
- Improve Electronic Records Management

These recommendations are focused on:

- Improving compliance with state records management requirements
- Increasing the efficiency and effectiveness of records management
- Reducing records management costs

5.1 Enhance Records Management Oversight and Leadership

5.1.1 Enhance Records Management Officer Roles & Responsibilities *[Priority 1]*

Define, expand, and document responsibilities for the Records Management Officer. Example responsibilities include the following:

- Establish goals for records management.
- Establish metrics for records management.
- Develop and lead improvement initiatives for records management.
- Develop and provide records management training in coordination with the Learning & Performance Center.
- Adjudicate and manage waiver requests for extending records retention periods – consider having the County Administrator or a delegated authority sign off on these waivers, as well.
- Provide advice and consultation to departments to support them in addressing audit findings.
- Provide advice and consultation to departments / offices in making records management process and performance improvements.

Defining and fulfilling these responsibilities will help the County's records management program to be more compliant, efficient, and effective.

The expansion of the Records Management Officer responsibilities will require an increased level of effort, particularly at the beginning. Consider allocating additional resources initially

and assess the steady-state level of effort required to operate this enhanced approach. Additional resources may be needed.

Enhancing Records Management Officer roles and responsibilities will help to address the following observations:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Hardcopy Retention Required for Signed Records
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- No RM-3 Form for Local Records Destruction
- Lack of Authority and Resources
- eRecord File Format Confusion
- Digitization Prioritization

5.1.2 Provide Training *[Priority 2]*

The Learning & Performance Center, in partnership with the Records Management Officer, to develop and provide Records Management training to Record Owners, Record Administrators, and employees. Robust training will help to address the following observations:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Hardcopy Retention Required for Signed Records
- Confusing Records Transfer Process
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- No RM-3 Form for Local Records Destruction

As part of the new training offerings:

- Consider developing asynchronous online training, so employees can take the training anytime. Consider establishing an annual requirement for all new employees to take records management training as part of their onboarding process.
- Consider establishing an annual requirement for all employees to take “refresher” records management training or including a records management slide as part of other annual County training.

Prioritize training for Record Liaisons, as they have the greatest potential impact into records management operations.

Next, prioritize training of employees in departments / offices that produce the most hardcopy records per month, as reported during IT Cadre interviews with their Records Owners and Records Liaisons:

- Accounting
- Utilities
- Commonwealth Attorney

5.1.3 Conduct Audits and Provide Support *[Priority 1]*

Chesterfield Internal Audit organization to audit departments / offices on their records management compliance and performance.

IST to provide advice and consultation to Departments to support them in addressing audit findings.

Chesterfield Internal Audit paired with the Records Management Officer must be granted the necessary authority to enforce compliance with records management requirements on the County departments.

Conducting audits and providing support will help to address the following observations:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Hardcopy Retention Required for Signed Records
- Confusing Records Transfer Process
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- No RM-3 Form for Local Records Destruction
- Lack of Records Management in Software Platforms

Items to be included in the audit scope include:

- Does the department / office have a complete and accurate Records Locator Inventory (RM-20 form)? Assist each department / office in developing and refining the inventory.
- Are records (electronic & hardcopy) being identified completely and correctly? Assist each department / office in distinguishing records from non-records.

- Are records (electronic & hardcopy) being assigned the correct retention schedule? Assist each department / office in selecting appropriate retention schedules. Prioritize validation of records tagged with permanent or other long retention periods.
- Are records (electronic & hardcopy) being destroyed in accordance with the assigned retention schedule? Advocate for new technologies and techniques to make records destruction easier and more consistent.
- Are waivers in place where retention periods are inappropriate? Assist departments / offices in pursuing waivers, as needed.
- Are there opportunities to digitize more records / manage fewer hardcopy records? Assist in identifying opportunities to digitize more records and to make the digitization process more effective.
- Are there opportunities to receive more native digital records to reduce the need for scanning? Assist departments / offices in identifying opportunities to reduce the need for scanning.
- Are there duplicate records being held within the department / office that can be eliminated (e.g., hardcopy and electronic copes of the same record)? Assist in identifying and mitigating the perceived need to manage duplicate records.
- Is the department's / office's Record Locator Inventory (RM20 form) complete and accurate? Assist in resolving any issues.
- Are there duplicate records being held by multiple departments / offices that can be eliminated (e.g., duplicative records held by the department / office and HR or Accounting)? Assist departments / offices in clarifying record ownership across departments / offices and eliminating duplicate records.

Prioritize audits of departments / offices that produce the most hardcopy records per month, as reported during IT Cadre interviews with their Records Owners and Records Liaisons:

- Accounting
- Commonwealth Attorney
- Utilities

5.1.4 Control Off-Site Storage Costs *[Priority 2]*

Provide enhanced guidance and requirements for off-site hardcopy records storage with the goal of reducing costs associated with off-site records storage.

Example guidance and requirements include:

- New hardcopy records with relatively short retention periods should be stored locally, as local storage space permits, and NOT sent to off-site storage.

- Whenever a box of records is retrieved from VRC, the contents of the box should be scanned and destroyed (after the quality of the digitized copy has been validated) – not returned to VRC, unless there is a legal requirement to keep the box contents as hardcopy.
- New hardcopy records that are assigned a retention period of greater than or equal to 15 years should be digitized, and the hardcopy should be destroyed (after the quality of the digitized copy has been validated), unless there is a legal requirement to keep the box contents as hardcopy. Calculations supporting this recommendation are shown in Appendix C.

Prioritize analysis of departments / offices that, based on actual records storage data from VRC:

- Have the most boxes stored at VRC
    - Mental Health Support Services
    - Police
    - Utilities
- Send the most boxes of records to VRC
    - Commonwealth Attorney
    - Police
    - Procurement
- Retrieve the most boxes of records from VRC
    - Commonwealth Attorney
    - Police
    - Procurement

Off-site storage is a critical part of records management within the County. To ensure optimization and to identify opportunities for improvement, conduct an assessment of VRC cost and performance:

- Total cost as compared to other vendors
    - Pickup cost
    - Storage cost
    - Retrieval cost
    - Destruction cost
- Performance as compared to other vendors
    - Box pickup and retrieval timeliness
    - Box inventory completeness and accuracy
    - Box inventory retrieval performance (lost boxes)
    - Box search / retrieval interface effectiveness (to search for and request retrieval of boxes)

Controlling off-site storage costs will help to address the following observations:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Successful Records Digitization & Destruction
- Hardcopy Retention Required for Signed Records
- Lost Records
- eRecord File Format Confusion
- Police RMs-LaserFiche Records Duplication
- Digitization Prioritization

5.1.5 Define and Track Measures & Metrics *[Priority 2]*

Collect data to provide insight into records management operations, performance, and effectiveness. Start with basic measurements and metrics that are easy to collect and analyze. As the records management program matures and more data is available, enhance the metrics to gain improved insight.

Example measures and metrics include:

- Breakout of the quantity of records being managed, by department / office and by type (e.g., hardcopy, electronic, on legal hold, in each retention schedule, due/overdue for destruction, stored off-site, stored on-site, duplicates)?
- Quantity of boxes retrieved from off-site storage, by department / office
- Records Management direct costs, by department / office (e.g., sending boxes to off-site storage, storing boxes off-site, retrieving boxes from off-site storage, destroying boxes stored off-site)
- Quantity of locally-stored records being destroyed, by department / office
- Cost of destroying locally-stored records
- Time spent scanning hardcopy records, by department / office
- Rate of change for off-site records storage (how quickly the quantity of boxes stored off-site is growing or shrinking)
- Audit results

Develop and publish metrics in the form of a scorecard for each department's / office's records management performance. Recognize high performers. Share substantial improvement efforts and results as success stories for other departments / offices to replicate. Include this as part of a records management awareness campaign (see recommendation, below).

Prioritize collection, analysis, and sharing of metrics related to VRC (e.g., boxes stored, new boxes sent, boxes retrieved). This data is relatively easy to obtain and provides some insight into records management progress, by department / office.

Defining and tracking metrics will help to address the following observations:

- Records Retained Beyond Retention Period
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction

5.1.6 Develop Enhanced Records Transfer Process and Instructions [QUICK WIN] *[Priority 2]*

The Records Management Officer should develop enhanced instructions and guidance for the records transfer process, to include clarifications to address commonly received questions or concerns from Record Administrators.

Investigate the possibility of developing a workflow-driven, self-service application for Record Administrators to send and retrieve records to/from off-site storage.

Developing an enhanced records transfer process and Instructions will help to address the following observations:

- Confusing Records Transfer Process
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- No RM-3 Form for Local Records Destruction

5.1.7 Conduct Awareness Campaign *[Priority 3]*

Plan and implement an awareness campaign to communicate the importance of records management and to incentivize compliance with records management requirements.

An effective Awareness Campaign will help to address the following observations:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Hardcopy Retention Required for Signed Records
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- No RM-3 Form for Local Records Destruction

The Awareness Campaign could include the following:

- The County Administrator to release a statement to employees sharing the importance of records management and the need for compliance.
- Consider including records management compliance in employee performance reviews.
- Develop and publish metrics in the form of a scorecard for each department's / office's records management performance. Recognize high performers. Share substantial improvement efforts and results as success stories for other departments / offices to replicate.
- Communicate the advantages of electronic records over hardcopy. Seek to inspire employee confidence that electronic records are just as secure and even more easily retrievable as compared to hardcopy records. In some case, trust in electronic records is low in some cases due to concerns such as changing file format requirements, loss of data when systems / vendors change, and the risk of losing a record with a single click.

5.1.8 Relocate Records Management Officer Authority *[Priority 3]*

Consider moving the Records Management Officer responsibility from its current department-level position to a County Administration-level position. The County Administration-level position, directly supporting the County Administrator, should have the authority and resources necessary to make and enforce strategic Records Management-related decisions. This addresses the observation: Lack of Authority and Resources.

In our discussion with Chad Owen, LVA Records Management Section Manager, on 09Dec24, when asked about the ideal candidate and placement of the Records Management Officer in an organization, his opinion was that the role should be filled by a compliance-oriented person in an Administration-level position.

As an example, when a department / office requests a waiver to extend the retention period of a record, this introduces risk to the County. The decision to accept that risk or to push-back on the request would be better aligned with a County Administration-level position, rather than department-level position.

5.2 Reduce the Quantity of Hardcopy Records

5.2.1 Investigate Perceived Need for Duplicate Records *[Priority 1]*

Assess department- / office-specific duplicate record-keeping rationale and seek to eliminate the need for duplicate records.

This addresses the observations:

- Maintaining Multiple Copies of Records
- Hardcopy Retention Required for Signed Records

Work with those departments / offices who, as reported during IT Cadre interviews with their Records Owners and Records Liaisons, keep duplicate records to identify the specific records that are duplicates, understand the specific reason(s) why the records are duplicated, and take action to eliminate the need for duplicate records.

Prioritize those departments / offices with the highest quantity of duplicate electronic / hardcopy records:

- Accounting
- Environmental Engineering
- Juvenile Justice Services
    - Uncertainty in determining which organization owns the "official" copy of a record
- Police
- Utilities

5.2.2 Review, Digitize, and Destroy Hardcopy Records *[Priority 2]*

Several departments / offices have conducted highly successful efforts to review their hardcopy records (stored locally and off-site), destroy any non-records that are mixed in with records, validate assigned retention schedules, digitize records as appropriate, destroy hardcopy records that were digitized (after the quality of the digitized copy has been validated), and send hardcopy records that were not digitized to off-site storage.

This recommendation addresses the observation:

- Maintaining Multiple Copies of Records
- Records Retained Beyond Retention Period
- Successful Records Digitization & Destruction
- Hardcopy Retention Required for Signed Records
- Lack of Retention Schedule Assignment
- Police RMs-LaserFiche Records Duplication
- Digitization Prioritization

- The County should consider allocating resources (money, staffing, technology) to departments / offices to encourage their hardcopy records review, digitization, and destruction.
- The County should consider requiring all departments / offices to review their records at VRC as well as records that are stored locally, with the goal of:
    - Removing any non-records from the boxes
    - Confirming retention schedule assignment and retention periods
    - Destroying any records whose destruction period has ended, after confirming that there are no legal holds

    - Digitizing records with long retention periods and destroying the hardcopies (after the quality of the digitized copy has been validated)
    - Storing records with short retention periods locally
- The Circuit Court Clerk has requested funding and is seeking vendor quotes to digitize tens of thousands of hardcopy records currently stored locally. There could be an opportunity to leverage their research and experience to acquire digitization support for other County departments / offices.

Prioritize departments / offices that have the most hardcopy stored locally and at VRC (based on actual VRC data), and departments / offices whose records are >90% hardcopy (as reported during IT Cadre interviews with their Records Owners and Records Liaisons):

- Most records stored locally
    - Community Corrections
    - Commonwealth Attorney
    - Emergency Communications
- Most records stored at VRC
    - Mental Health Support Services
    - Police
    - Utilities
- >90% hardcopy records
    - Commonwealth Attorney
    - Community Corrections
    - Environmental Engineering
    - General Services
    - Human Resources
    - Libraries
    - Real Estate Assessments
    - Police
    - Treasurer

For departments that keep duplicate hardcopy and electronic records (as reported during IT Cadre interviews with their Records Owners and Records Liaisons), the hardcopy records can be destroyed (after the quality of the digitized copy has been validated).

- Environmental Engineering
- Police
- Accounting
- Utilities
- Board of Supervisors

5.2.3 Investigate Wet Signature Record Requirements *[Priority 2]*

Work with the County Attorney to seek definitive permission to digitize and destroy wet signature records to the greatest extent possible. Document these decisions and direct departments / offices to take action.

The Code of Virginia §42.1-86.01 authorizes the retention of electronic records in place of hardcopy. In our discussion with Chad Owen, LVA Records Management Section Manager, on 09Dec24, when asked about wet signatures, he confirmed that from the State of Virginia's perspective, hardcopy wet signatures are not required… a signed document that is digitized can serve as the record with no need to retain the hardcopy. However, other bodies, such as the Federal Government, may have wet signature requirements that supersede Virginia requirements.

This recommendation addresses the observations:

- Hardcopy Retention Required for Signed Records
- Successful Records Digitization & Destruction
- Hardcopy Retention Required for Signed Records

While this issue likely impacts most departments / offices, prioritize departments / offices that reported specific concerns about this issue during IT Cadre interviews with their Records Owners and Records Liaisons include:

- Community Enhancement
- Learning & Performance
- Parks & Recreation
- Planning
- Procurement
- Real Estate Assessments

5.2.4 Define Electronic Record Format Requirements [QUICK WIN] *[Priority 2]*

Research best practices for electronic record formats (e.g., PDF, TIFF), select and document the preferred and acceptable formats, and provide direction to departments / offices. This recommendation addresses the observation: eRecord File Format Confusion.

See Appendix D for a discussion of this topic.

5.2.5 Exchange More Electronic Records, Fewer Hardcopy *[Priority 2]*

Work with external stakeholders who currently receive hardcopy records from Chesterfield County to accept electronic records. Hardcopy will be required in some cases, but it should be

minimized. This will eliminate the need for the County to keep these records as hardcopy or to print hardcopies of electronic records.

Work with external stakeholders who currently provide hardcopy records to Chesterfield County to, instead, provide those records electronically. Hardcopy will be required in some cases, but it should be minimized. This will eliminate the need for the County to digitize these records and/or eliminate the cost to store and manage hardcopy records.

Prioritize departments / offices that reported the highest level of effort dedicated to records management during IT Cadre interviews with their Records Owners and Records Liaisons, as this measure is a proxy for departments / offices that do the most scanning of hardcopy records:

- Utilities
- Human Resource Management
    - Pain point: manually uploading annual employee review records into LaserFiche, one-at-a-time)
- General Services

Exchanging more electronic records / fewer hardcopy records will help to address the following observation: Successful Records Digitization & Destruction.

5.3 Comply with LVA Records Retention Requirements

5.3.1 Develop Records Locator Inventory (LVA RM-20 Form) *[Priority 1]*

Departments / offices to fill out the Records Locator Inventory form (LVA RM-20 Form) for their department, which includes hardcopy and electronic records. This inventory is required by LVA. The process of developing this inventory form will reveal weaknesses in their records management processes and opportunities for improvement. Maintaining a complete and accurate Records Locator Inventory, documented in the RM-20 form, is an LVA requirement.

Departments developing their Record Locator Inventories (RM-20 forms) will help to address the following observations:

- Lack of Separation for Records vs. Non-Records
- Lack of Retention Schedule Assignment

5.3.2 Manage LVA Records Retention Schedule Deviations *[Priority 2]*

Several departments / offices report that some of their records match a given LVA record description, but the prescribed retention period is inappropriate (typically it is viewed as being too short). In some cases, records are subject to retention period requirements from other organizations (e.g., the Federal government) that conflict with the LVA retention periods.

Keeping records beyond the LVA retention schedule results in non-compliance with LVA requirements. There are no waivers available to LVA records retention schedules.

To ensure compliance, the County should consider taking the following actions:

- The Records Management Officer should establish a formal process to review and adjudicate requests to keep records beyond the retention schedule. Document decisions and revisit when appropriate.
- Find a Better Retention Schedule / Record Series#: In coordination with the Records Management Officer, the department / office seeking to keep a record beyond the retention period should review all available retention schedules to see if the record could be assigned a different retention schedule / record series # that has the desired retention period.
- Pursue a Change to the Retention Schedule / Record Series#: Record retention scheduled are developed by LVA in coordination with all subject organizations across the state of Virginia. If a change to a retention period is desired, there is a substantial Virginia Records Oversight Committee process to follow. Any changes to retention schedules are applicable to all Virginia state organizations – not just the requestor's organization. The department, in coordination with the Records Management Officer, could work with LVA to pursue a retention schedule / record series# change.

There are no fines or legal penalties associated with keeping records beyond their retention period. However, there is a cost and legal exposure associated with storing and maintaining records for a longer period than necessary. Consider including the County Administrator or a designee in the decision to keep a record beyond its retention period.

This recommendation addresses part of observation: Records Retained Beyond Retention Period.

Several departments / offices manage case files that must be kept together, which include records with multiple retention schedules and retention periods. The Records Management Officer to define and publish policy for how to handle this situation in a compliant way.

Prioritize departments / offices who report this issue during IT Cadre interviews with their Records Owners and Records Liaisons:

- Accounting
- Budget & Management
- Building Inspection
- Community Engagement
- Community Enhancement
- Environmental Engineering
- Learning & Performance Center
- Fire-EMS
- General Services
- Juvenile Justice
- Mental Health Support Services
- Police
- Police – Animal Control
- Real Estate Assessments
- Sheriff
- Social Services
- Transportation
- Utilities

5.3.3 Add Box# to LVA Records Destruction Workflow [QUICK WIN] *[Priority 2]*

Only the record total weight/volume, instead of box quantity/barcode number, appears on the LVA workflow "Record Owner" task. The Record Owner must approve the workflow task to move forward, but without any positive confirmation of which specific records they are designating for destruction.

To resolve this issue, the Records Management Officer should leverage one of the existing free text fields in the LVA RM-3 workflow (e.g., "Location" or "Record Series Notes" field) to include the barcode numbers for the boxes of records to be destroyed.

This recommendation addresses the observation: Inability to Confirm Correct Records Destruction.

5.4 Improve Electronic Records Management

5.4.1 Enhance County-wide Electronic Records Management Policy *[Priority 1]*

IST to enhance existing records management policy (Policy #7-12) to include additional guidance and requirements for managing electronic records. This policy will be developed with consideration of the other applicable records management-related governance documents that are shown in Appendix G. Ensure that all departments are given the opportunity to review and comment on the policy. Example content includes the following items:

- Destroy the hardcopy after digitizing it and validating the quality of the digitized file

- Work with Legal to determine which records must be kept as hardcopy (e.g., certain wet-signature documents) and provide documented direction to departments / offices
- Electronic records must be identified / flagged as records
- Electronic records must be assigned a retention schedule
- Retention periods for electronic records must be tracked – departments / offices must be able to generate a report of electronic records with expired or expiring retention periods
- Electronic records must be destroyed in accordance with the assigned retention schedule, unless a legal hold or other waiver is in place
- Complete an RM-3 form when destroying digitized records
- Departments / offices to define a standard folder structure and file naming convention for their records
- Departments / offices to define a records management plan for each of their software platforms that house records
    - How to identify records in the system
    - Approach for assigning retention schedules to folders, documents, and data
    - Approach for deleting records when their retention period ends

- Acceptable and recommended file format(s) for electronic records (e.g., PDF, TIFF)
- Guidance and requirements for keeping hardcopy vs digitizing (e.g., digitize all records with retention period >=15 years)
- Guidance for adopting Digital Rights Management and Data Loss Prevention tools, which will leverage artifact tagging to assign to a record class and associate with the appropriate retention policy.
    - As the County considers moving to Microsoft M365 G5 license entitlement level with the impending renewal of our MEA, it will gain access to much of this enhanced capability, including auto tagging.

Effective Electronic Records Management Policy will help to address the following observations:

- Maintaining Multiple Copies of Records
- Lack of Separation for Records vs. Non-Records
- Records Retained Beyond Retention Period
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- Lack of Records Management in Software Platforms
- eRecord File Format Confusion
- Digitization Prioritization

5.4.2 Define Records Management Plans for Software Platforms *[Priority 2]*

Departments / offices to define a records management plan for each of their software platforms that house records. For folder-based software platforms (e.g., network drives, SharePoint), these plans may include an approach to tagging records with retentions schedules and/or mapping standardized folder structures and file naming conventions to retention schedules. Consider completing a blanket / annual RM-3 form for automated deletion of electronic records (e.g., automated monthly deletion of security camera footage).

This recommendation addresses the observations:

- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- Lack of Records Management in Software Platforms

The plan / folder structure and file naming must be sufficient to:

- Easily identify a record vs a non-record
- Easily connect / assign record retention schedules to files
- Easily determine / forecast file retention periods and destruction target dates
- Easily find files whose retention periods have expired, in an automated fashion
- Identify records with a legal hold or other reason to delay destruction
- Easily destroy expired records, in an automated fashion

Prioritize the most heavily used software platforms:

- LaserFiche
- SharePoint
- Accela / ELM

5.4.3 Pursue Automated Retention Schedule Assignment and Records Destruction *[Priority 2]*

Departments / offices to define automated / semi-automated mechanisms to assign retention schedules & to destroy expired records stored in all software platforms that are owned / selected by Chesterfield County.

This may include the adoption of Digital Rights Management and Data Loss Prevention tools, which will leverage artifact tagging to assign to a record class and associate with the appropriate retention policy. Moving to Microsoft M365 G5 license entitlement level would gain access to much of this enhanced capability, including auto tagging.

Prioritize the most heavily used software platforms:

- LaserFiche
- SharePoint
- Accela

This action will help to address the following observations:

- Lack of Separation for Records vs. Non-Records
- Records Retained Beyond Retention Period
- Lack of Retention Schedule Assignment
- Lack of eRecord Retention Period Tracking
- Lack of eRecord Destruction
- Lack of Records Management in Software Platforms

6. Prioritization

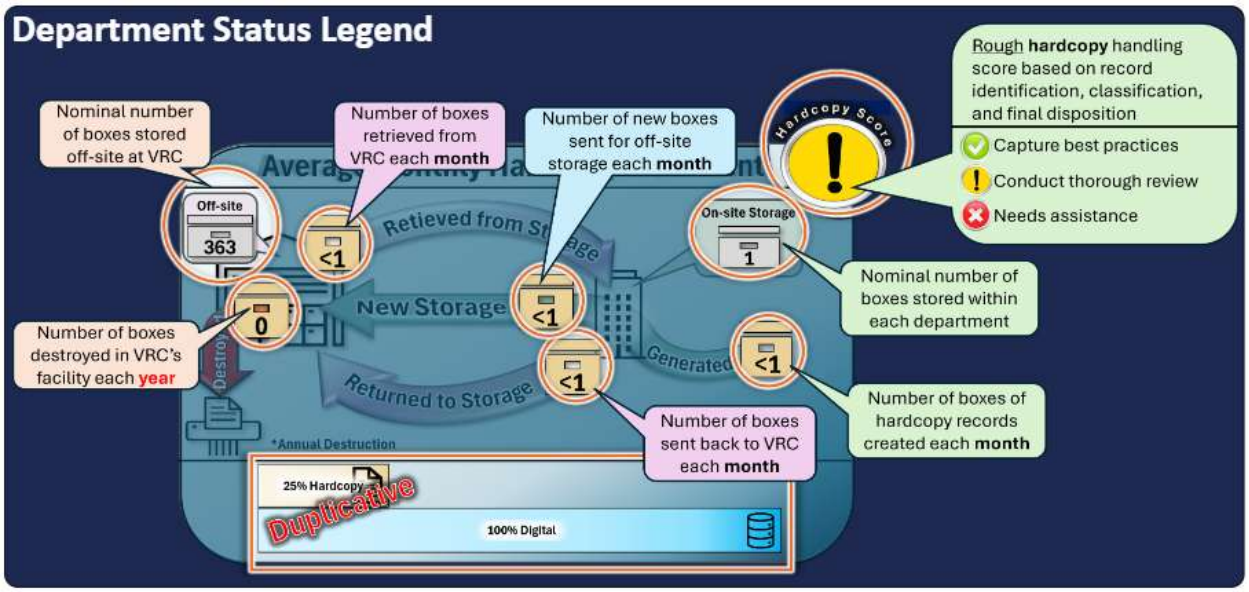
The table, below, lists the recommendations in order of priority. The priority is informed by: (1) the impact that implementing the recommendation will have on Chesterfield County's records management activities, and (2) the ease of implementation.

- A lower Impact ranking indicates a relatively more significant impact on Chesterfield County's records management activities (1 being the most significant impact)
- A lower Ease ranking indicates a relatively lower level of effort to implement the recommendation (1 being the easiest to implement)
- A lower Priority ranking indicates a relatively higher priority (1 being the highest priority)

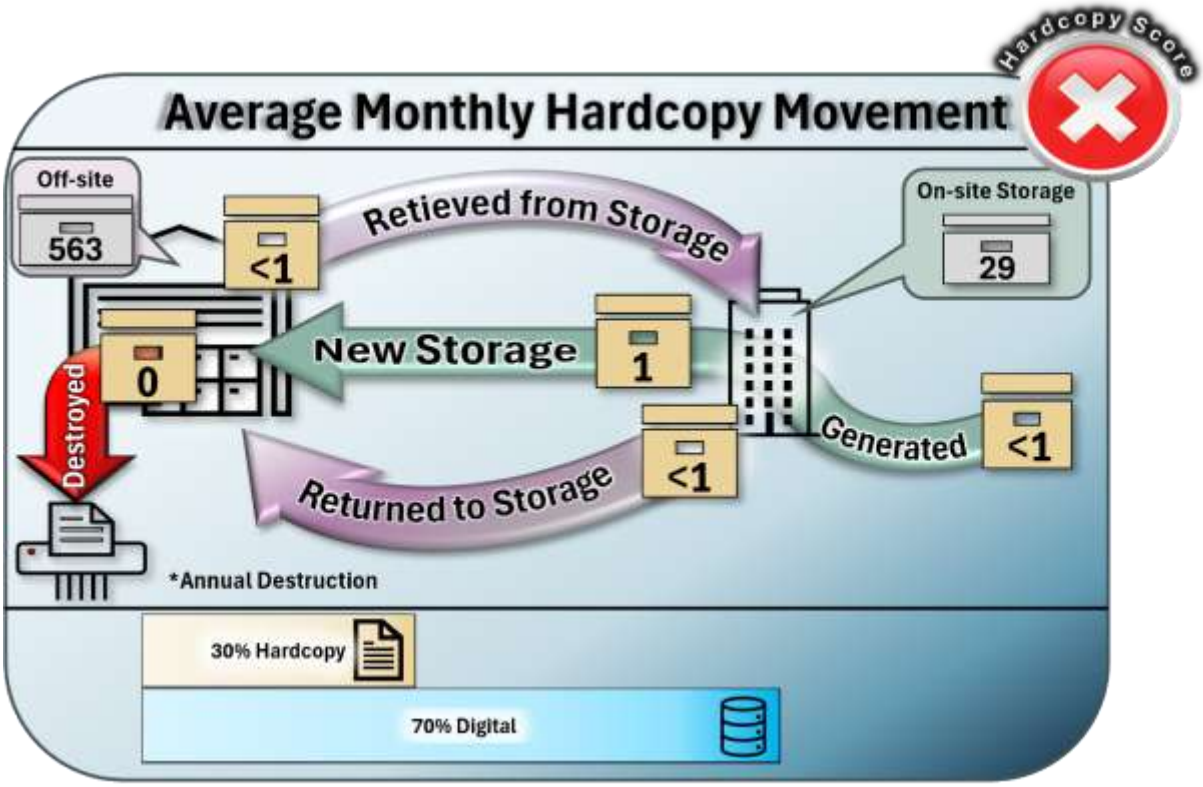
Priority	Impact	Ease	Group	Rec #	Recommendation
1	1	1	Enhance Records Mgmt Oversight & Leadership	1.1	Enhance Records Management Officer Roles & Responsibilities
	1	2	Reduce the Quantity of hardcopy Records	2.1	Investigate Perceived Need for Duplicate Records
	1	2	Improve Electronic Records Management	4.1	Enhance County-wide Electronic Records Mgmt Policy
	1	3	Enhance Records Mgmt Oversight & Leadership	1.3	Conduct Audits and Provide Support
	1	3	Comply with LVA Records Retention Requirements	3.1	Develop Records Locator Inventory (LVA RM-20 Form)
2	1	3	Reduce the Quantity of hardcopy Records	2.5	Exchange More Electronic Records, Fewer Hardcopy
	1	3	Improve Electronic Records Management	4.2	Define Records Mgmt Plans for Software Platforms
	1	3	Improve Electronic Records Management	4.4	Pursue Automated Retention Schedule Assignment and Records Destruction
	2	2	Enhance Records Mgmt Oversight & Leadership	1.2	Provide Training
	2	2	Reduce the Quantity of hardcopy Records	2.3	Investigate Wet Signature Record Requirements
	2	2	Comply with LVA Records Retention Requirements	3.2	Manage LVA Records Retention Schedule Deviations
	2	3	Enhance Records Mgmt Oversight & Leadership	1.5	Define and Track Measures and Metrics
	2	3	Reduce the Quantity of hardcopy Records	2.2	Review, Digitize, and Destroy Hardcopy Records
	3	1	Enhance Records Mgmt Oversight & Leadership	1.4	Control Off-Site Storage Costs
	3	1	Enhance Records Mgmt Oversight & Leadership	1.6	Develop Enhanced Records Transfer Process and Instructions
	3	1	Reduce the Quantity of hardcopy Records	2.4	Define Electronic Record Format Requirements
	3	1	Comply with LVA Records Retention Requirements	3.3	Add Box# to LVA Records Destruction Workflow
3	3	2	Enhance Records Mgmt Oversight & Leadership	1.8	Relocate Records Management Officer Authority
	3	2	Enhance Records Mgmt Oversight & Leadership	1.7	Conduct Awareness Campaign

7. Department-Specific Notes / Highlights

This section provides department- / office-specific records management information. At the top of each subsection, a summary graphic of the department's / offices' hardcopy records is provided. The legend graphic, below, describes the contents of these summary graphics.

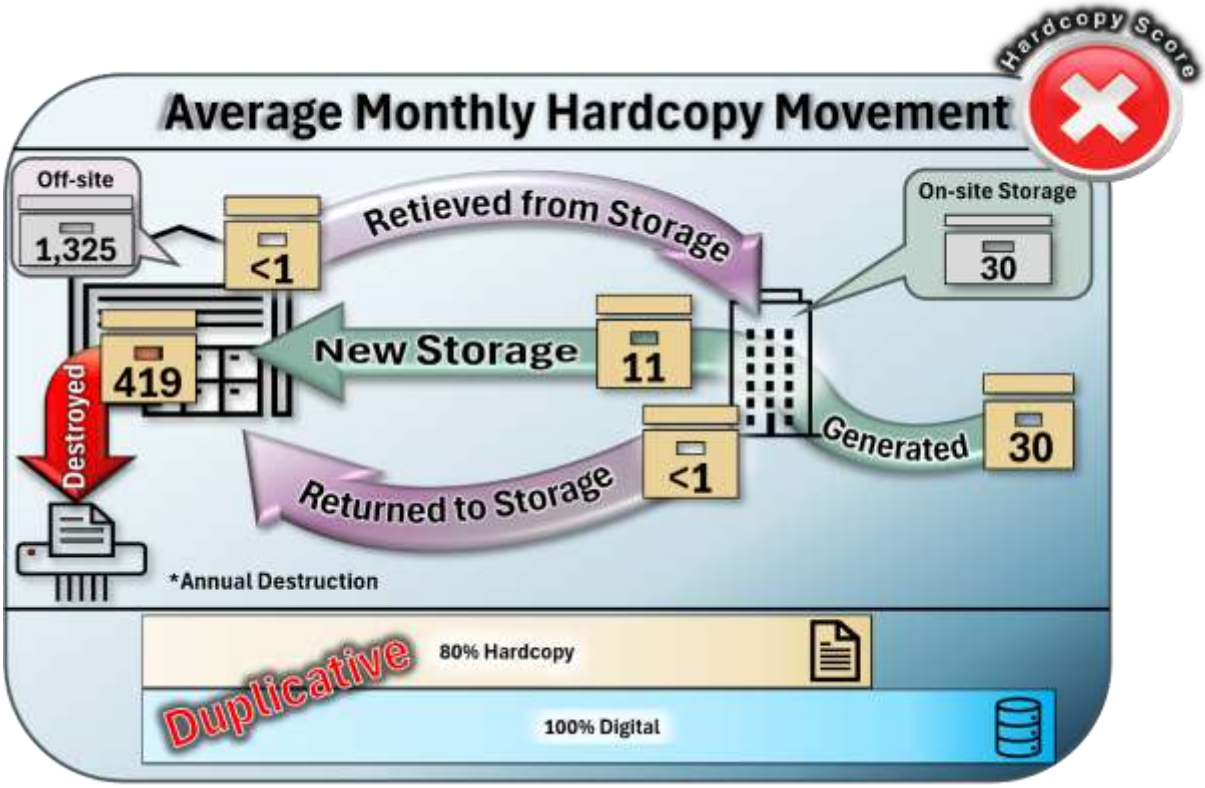


7.1 Administration



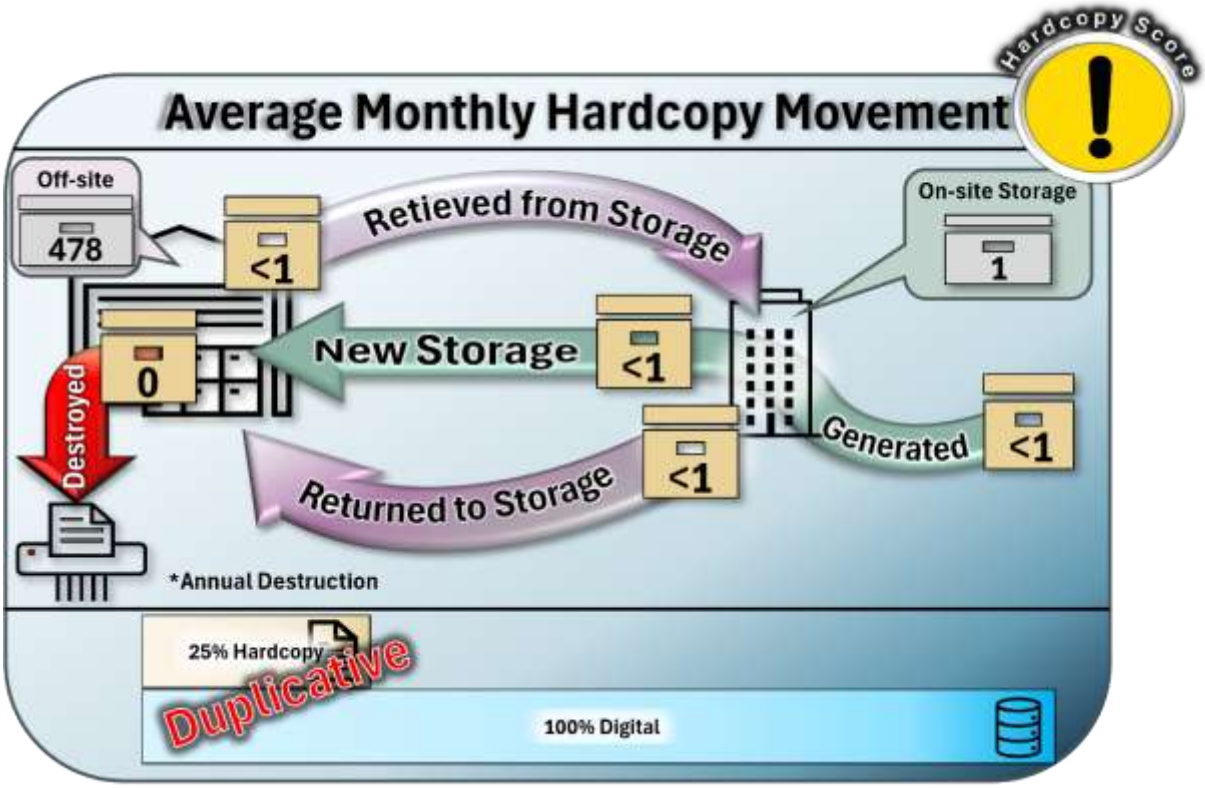
- 70% electronic / 30% hardcopy records
- All records are assigned Permanent retention
- Many records are kept as permanent, but are not associated with a specific LVA records retention schedule
- Policy Memo says to keep County Administrator records permanently

7.2 Accounting



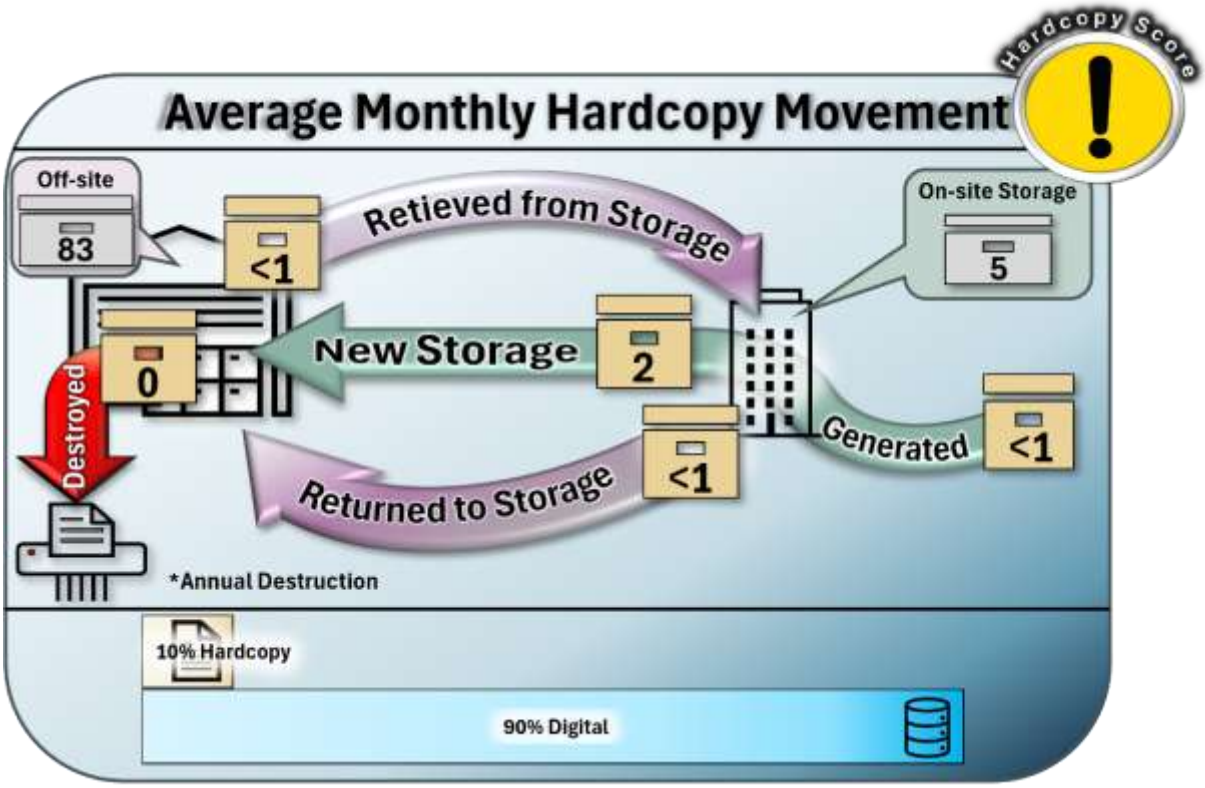
- Financial Systems and General Accounting = 100% electronic
- Accounts Payable (AP) and Payroll are 50% electronic.
- AP scans all invoices, but they have paper copies of 80% of those invoices. AP considers both versions to be records.
- No process for assigning a retention schedule or for destroying electronic records

7.3 Board of Supervisors



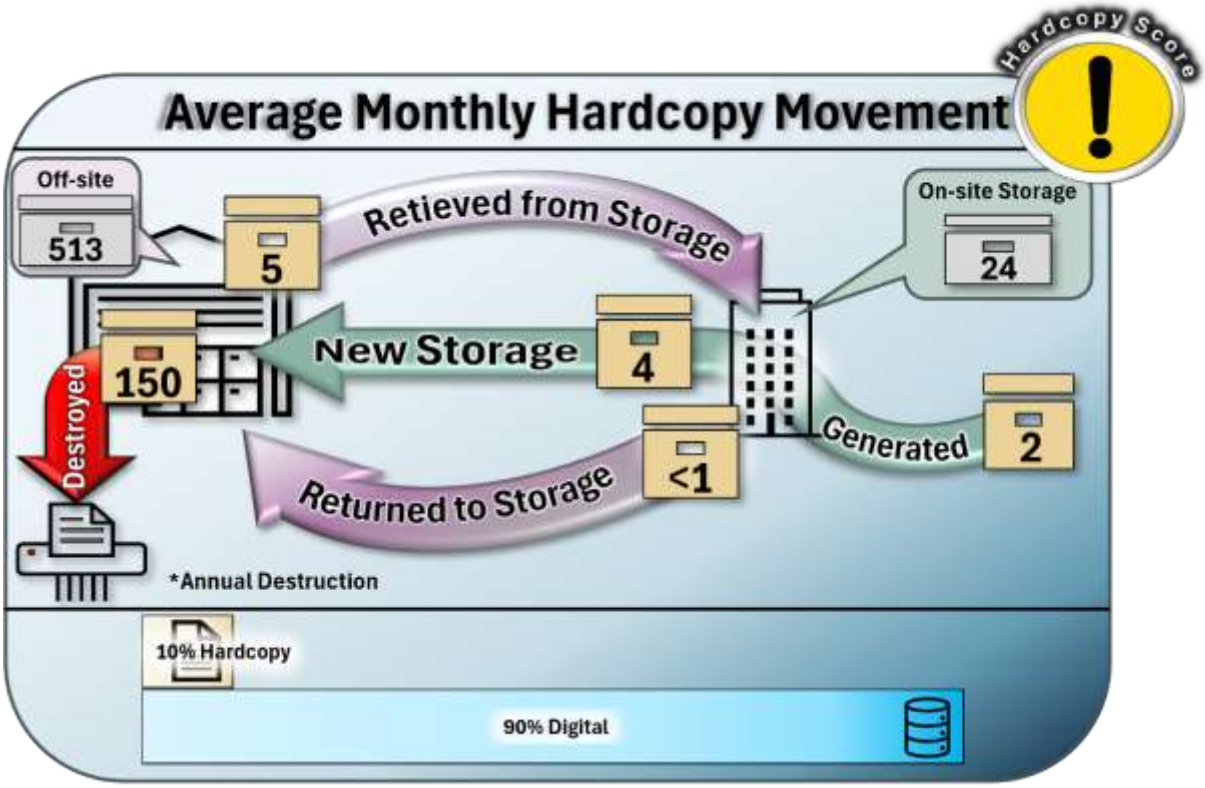
- 100% electronic / 25% hardcopy records
- Keep minutes in hardcopy and electronic formats
- No RM-3 form for destroying locally stored records
- No process for assigning a retention schedule or for destroying electronic records

7.4 Budget & Management



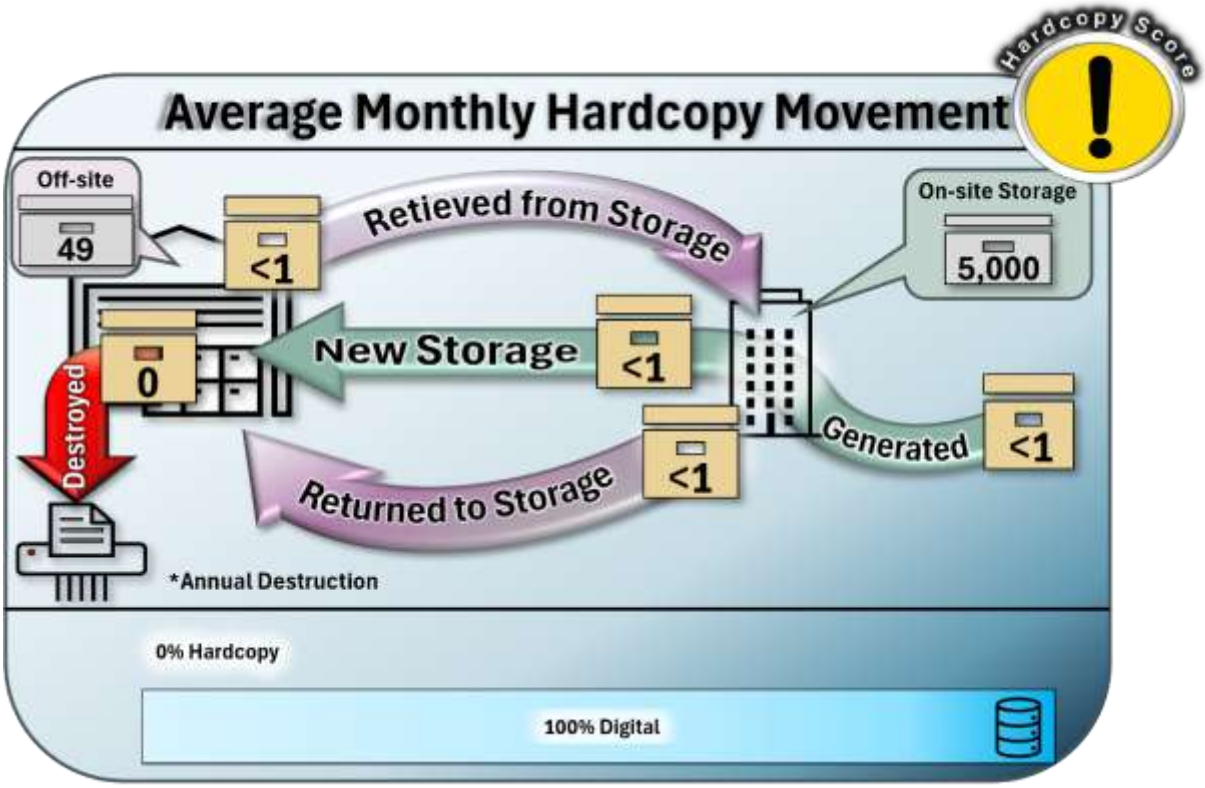
- 90% electronic / 10% hardcopy records
- Just did a major files cleanup – sent 19 boxes to VRC. Normal rate is <1 box per year
- They keep all records permanently… which is longer than the retention schedule requires
- Very few types of records – all are easily identified on SharePoint
- They would prefer to go to 100% electronic, but there is uncertainty about what electronic record format(s) are acceptable (e.g., PDF vs. TIFF)
- Recent document cleanout in mid-/late 2024

7.5 Building Inspections



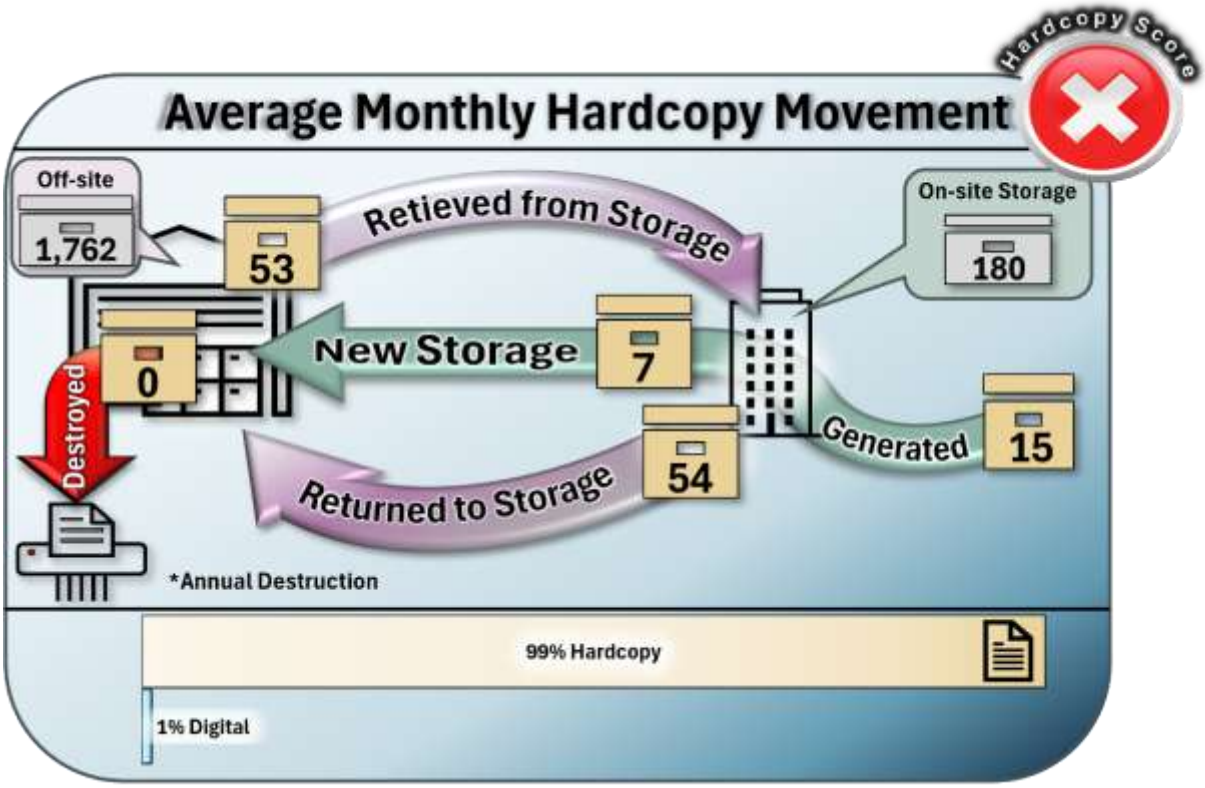
- 90% electronic / 10% hardcopy records
- No process for assigning a retention schedule or for destroying electronic records
- They would benefit from a review of their hardcopy work products to ensure that all records are identified, assigned a retention schedule, and boxed appropriately
- Accela record data and Accela document records housed in Laserfiche are over their 3-year retention limit, but they have not been destroyed
- Most commercial submissions arrive electronically. Residential are more likely to be hardcopy, although they are mostly electronic as well. The department has no plans to force citizens to submit via electronic format, due to concerns about small orgs not being able to do so. However, they do encourage electronic submissions highlighting faster handling than paper.
- When Accela system disruptions occur, submissions come in via hardcopy, which causes extra administrative effort.
- They are planning a major cleanout of the file room in November, prior to their office move.
- They keep hardcopy records, even after they scan them. They will consider changing this practice, but there is uncertainty concerning the rules associated with destroying hardcopy, even after scanning.
- They sometimes combine records with different retention periods in the same box for convenience and assign the longer retention schedule to the entire box.

7.6 Circuit Court Clerk



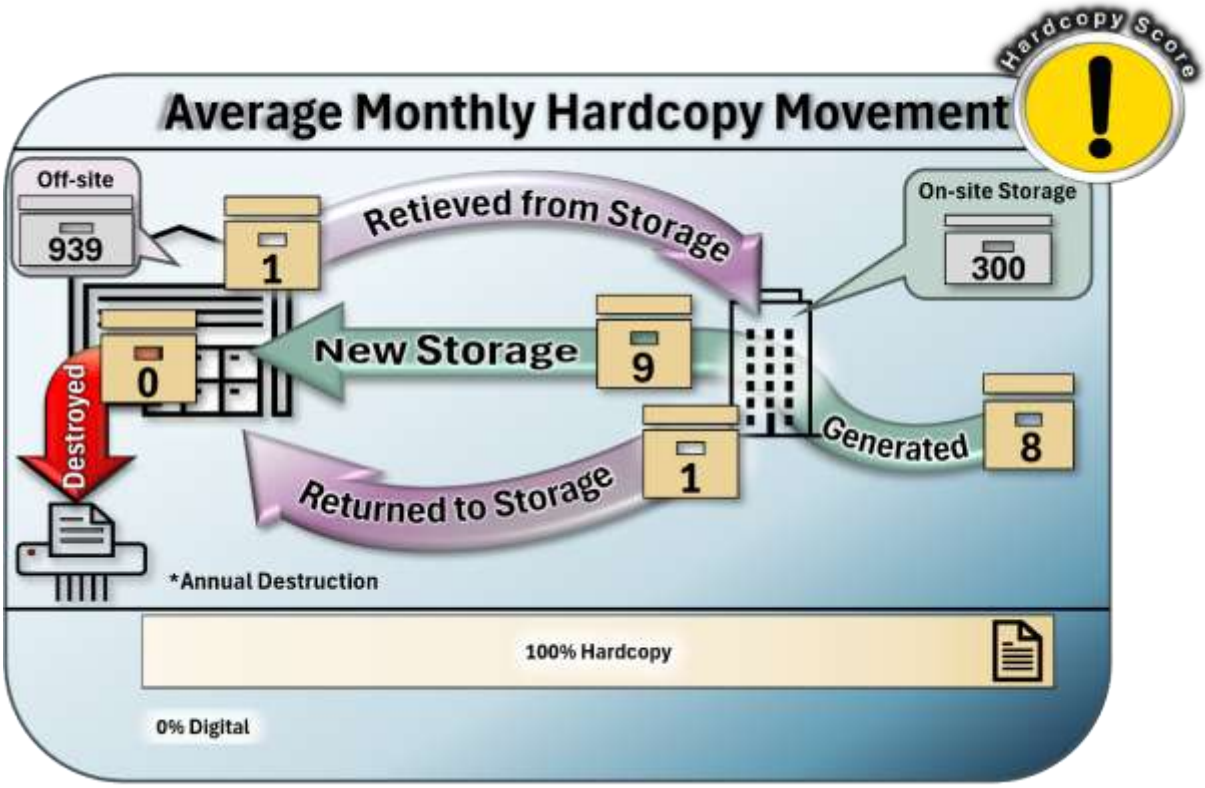
- 10 FTEs performing records management
- Moving to 99% electronic records on Jan 1, 2024, benefitted by the recent launch of the Digital Evidence Management platform. Only hardcopies will be those mandated to be hardcopy by law.
- Massive hardcopy storage problem. The basement of the building is full of records in "high density storage"
- They interpret the law to require them to store records locally – not at VRC
- They don't assign retention schedules to most records, other than election records and land records.
- They digitized all land records recently. They are going to digitize criminal, civil, and other records – funding has been requested and quotes are being sought.
- They use the State Supreme Court Office of the Executive Secretary, Circuit Information Systems, Circuit Case Management System, to store records. From the Clerk's perspective, records retention and destruction are handled by the State Supreme Court Office of the Executive Secretary.

7.7 Commonwealth's Attorney



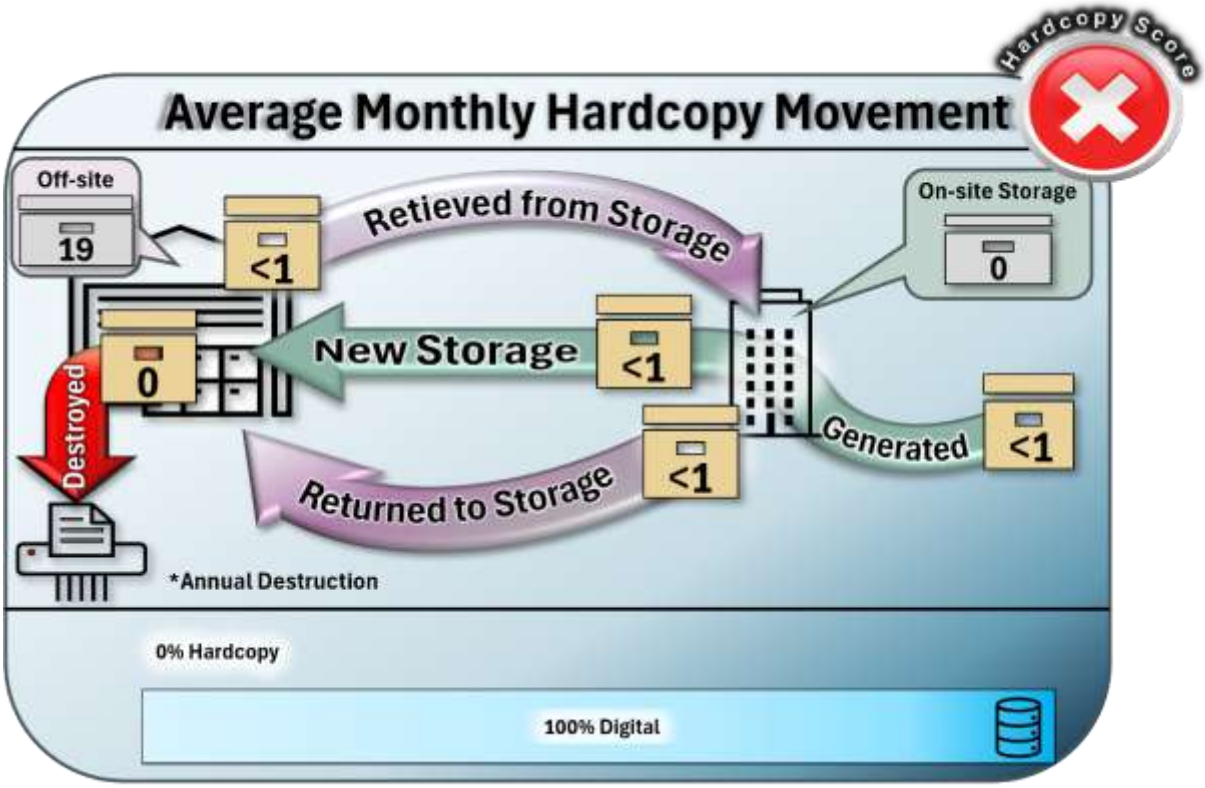
- 100% hardcopy
- No process for assigning a retention schedule or for destroying electronic records
- They use hardcopy primarily because of the difficulty of identifying and deleting electronic records
- No RM-3 form for destroying locally stored records
- The Commonwealth Attorney office is implementing a digital case evidence management solution in partnership with the County Police Department, targeted for release in late Spring 2025.

7.8 Community Corrections



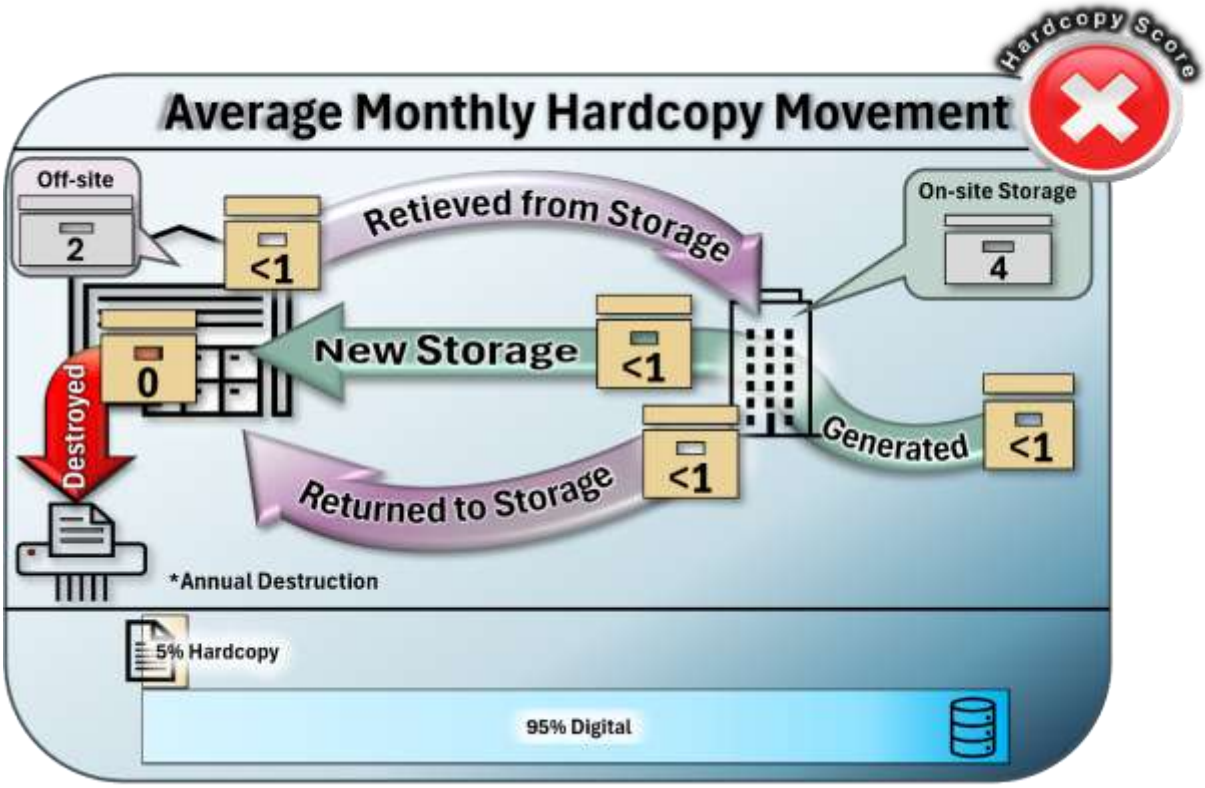
- 100% hardcopy records
- Records stay on-site in locked file cabinets for ~3 years (HIPAA requirement + quick access to these files is needed), until they are boxed and shipped to VRC, 1x per year
- No process for assigning a retention schedule or for destroying electronic records

7.9 Community Engagement & Resources



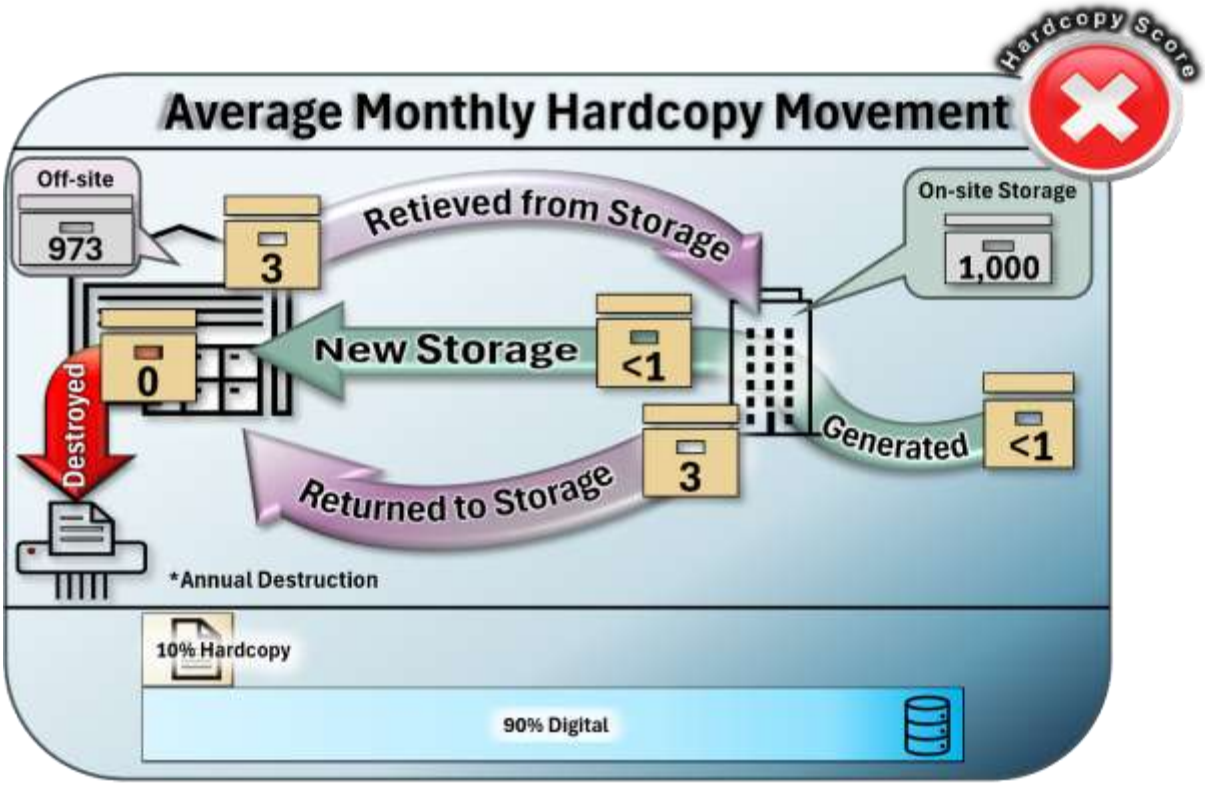
- 100% electronic
- Do not identify "records" – they keep all work products
- Never destroy any records
- Store electronic records side-by-side with non-records
- No process for assigning a retention schedule or for destroying electronic records
- Send no records to VRC and have no records stored at VRC
- They would benefit from having a standard folder structure to use to organize their files in SharePoint and V-Drive. If someone is out-of-office, it can be difficult to find a records in their unique folder structure.

7.10 Community Enhancement



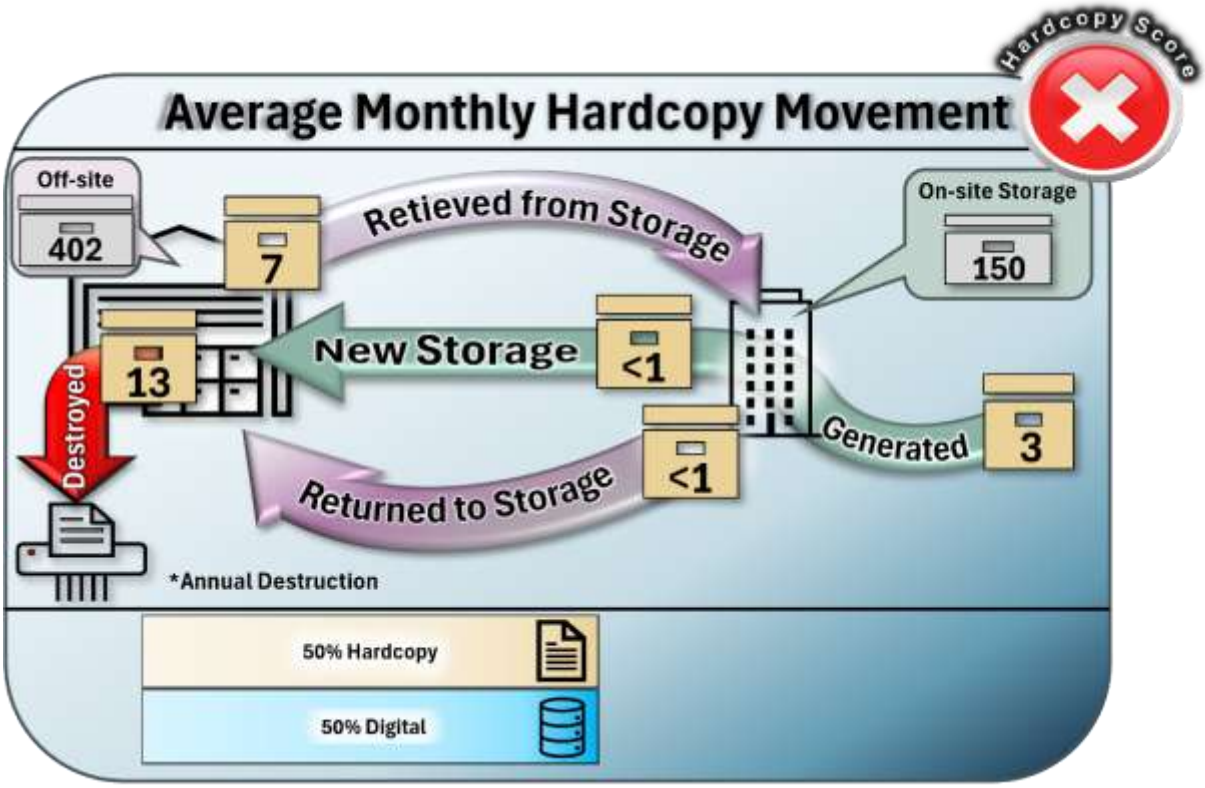
- 90% electronic / 10% hardcopy records
- Records stored on SharePoint, CE-Drive, and laptops – would benefit from an enhanced file storage plan
- No RM-3 form for destroying locally stored records
- No process for assigning a retention schedule or for destroying electronic records
- Hasn't sent a box to VRC is 3+ years
- Employees keep case files indefinitely for reference

7.11 County Attorney



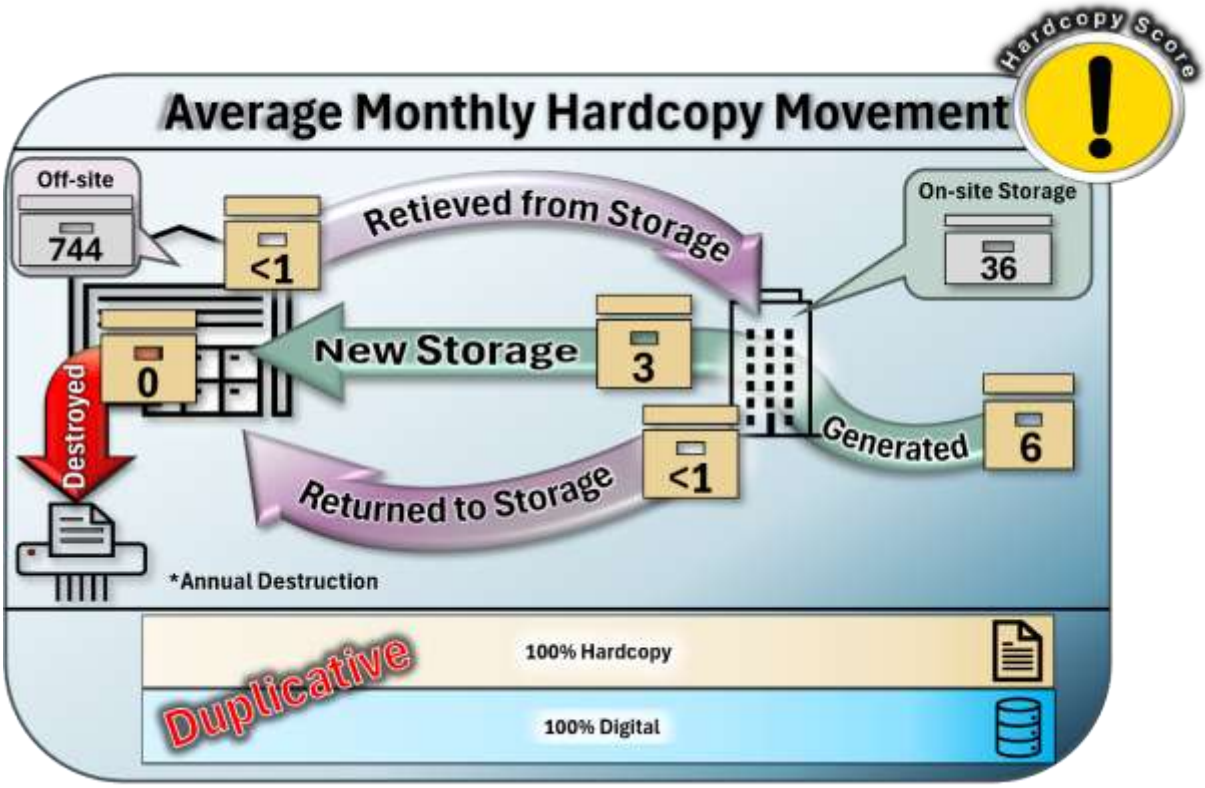
- 90% electronic / 10% hardcopy records
- They consider all work products to be records with permanent retention. No LVA schedule assigned.
- No process for assigning a retention schedule or for destroying electronic records
- Tag folders in Laserfiche to map to VRC or warehouse box numbers
- 1000 boxes stored in local warehouse, in addition to VRC boxes
- They would be open to digitizing warehouse & VRC records, but the quotes they received were way to high.

7.12 Emergency Communications



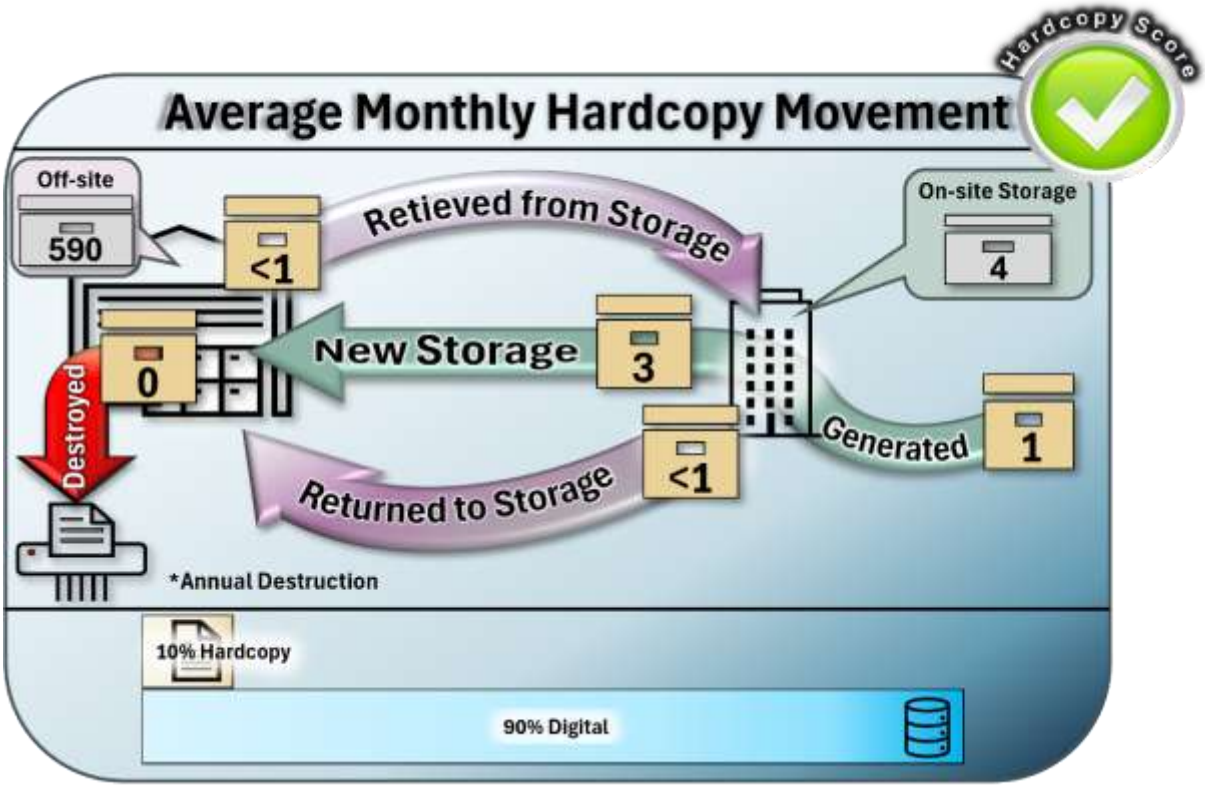
- 50% electronic / 50% hardcopy records
- 1x per year they go through the file cabinets, remove records that no longer need to be kept, and destroy them
- They auto-destroy 911 calls and radio calls with no RM-3 form. Perhaps they could get a blanket RM-3 form to destroy a year's worth of calls?
- No RM-3 form for destroying locally stored records
- No process for assigning a retention schedule or for destroying electronic records
- Uncertainty in who owns the master copy of some records (e.g., Police, Accounting and HR-related records)

7.13 Environmental Engineering



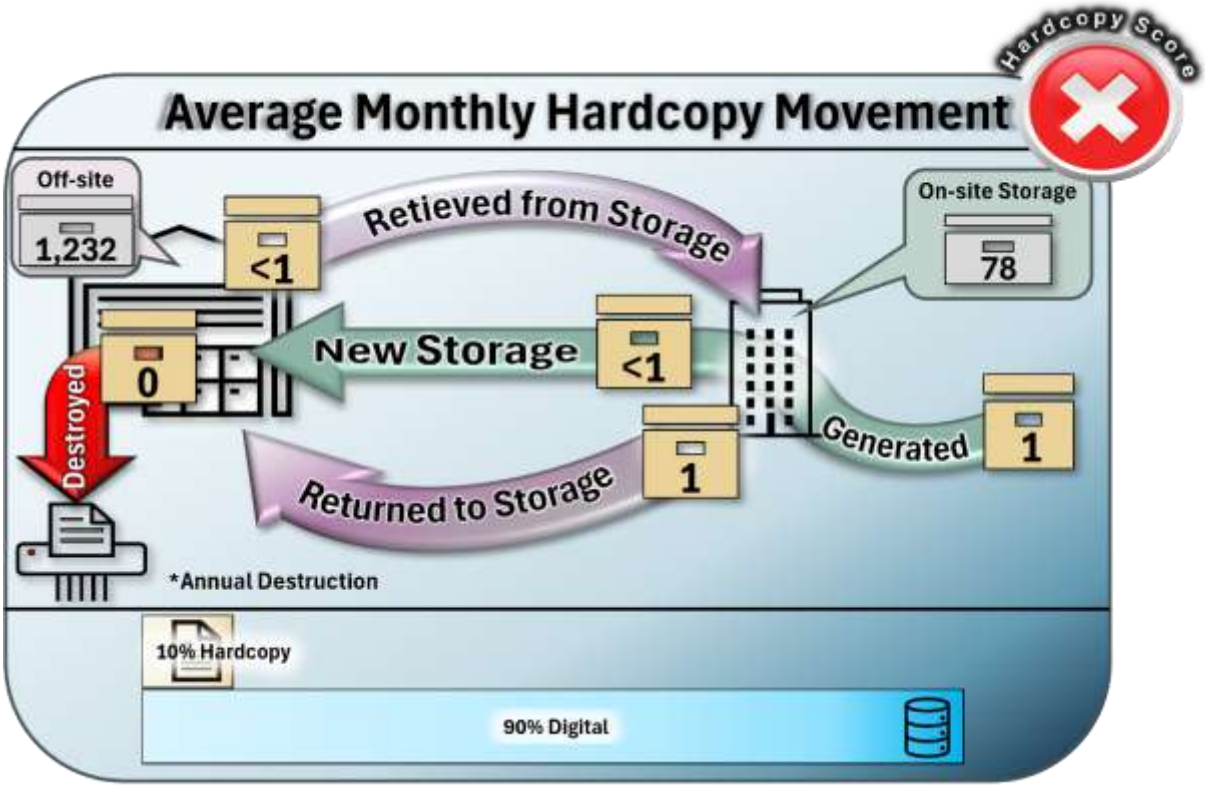
- 100% electronic records and 100% hardcopy records – considering both formats as records
- No process for assigning a retention schedule or for destroying electronic records
- They would prefer to go to 100% electronic, but there is uncertainty about what electronic record format(s) are acceptable (e.g., PDF vs. TIFF)
- Records with 1- to 3-year retention periods are kept locally, not sent to VRC
- No RM-3 form for destroying locally stored records
- When Mary sends list of VRC boxes for destruction, they retrieve the box from VRC, validate the contents, and destroy it themselves. This is because there are other retention requirements that may be applicable to the records in the box.
- They are moving toward having more electronic records and fewer digital
- They have hardcopy records that need to stay as hardcopy: large drawings – there needs to be a hardcopy on the construction site and inspectors prefer to bring hardcopy with them. Also, they issue hardcopy stamped plans that are used on construction sites. They often review multiple hardcopy drawings side-by-side. This is difficult to do in electronic form.
- They have a backlog for scanning. There are busy times and mellow times. They are staffed for the mellow times. They wish they had a service to provide scanning surge capacity during the busy times.
- It is very rare for them to pull hardcopys from VRC – only for FOIA requests and lawsuits
- They have records that are kept beyond the LVA retention period, because they are needed for reference

7.14 Fire-EMS



- 95% electronic / 5% hardcopy records
- Hardcopy records are moved offsite after a year or so
- Most of their records have Permanent retention
- No process for assigning a retention schedule or for destroying electronic records
- Vector Solutions houses new and old records, but only the new records are searchable

7.15 General Services



7.15.1 General Services / Capital Projects

- 100% electronic
- All records are kept permanently
- No records destruction
- No specific retention schedule assigned to new records, which are kept as Permanent
- Electronic records organized in easily navigable folder structure

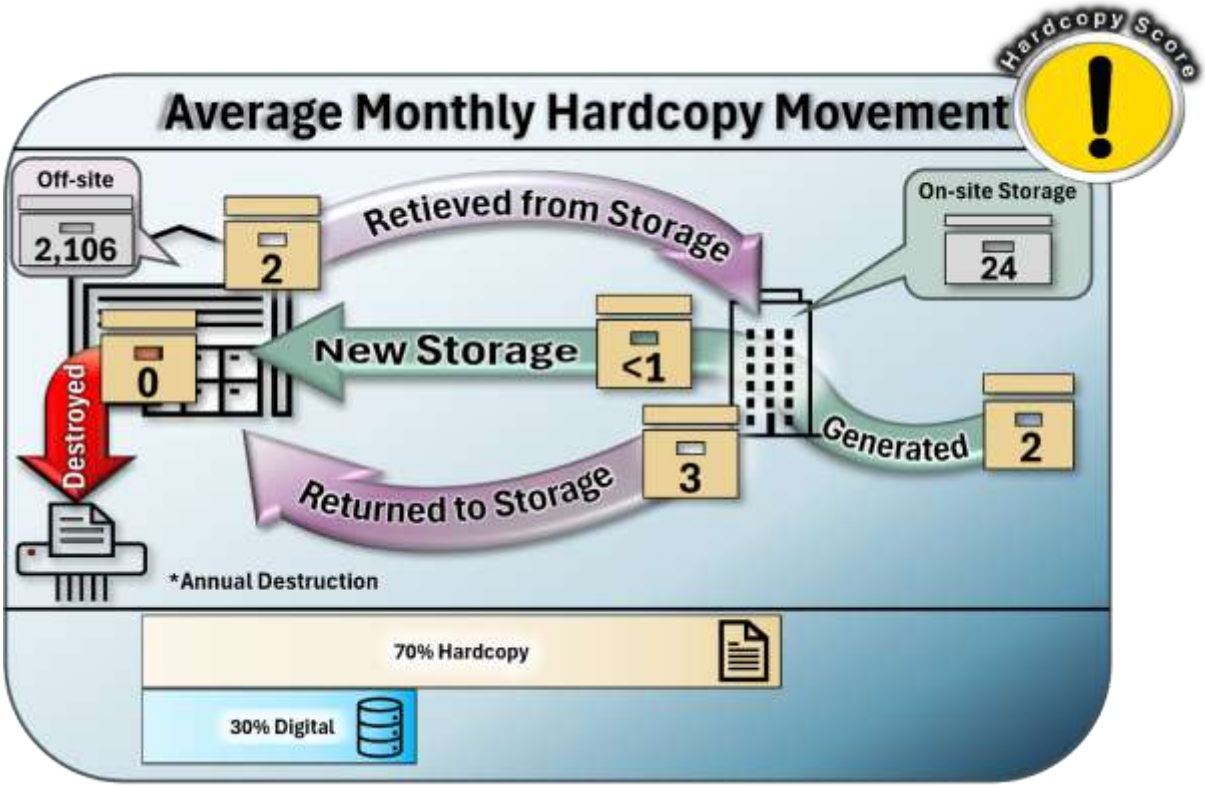
7.15.2 General Services / Fleet Management

- 100% electronic records (except for hardcopy invoices, which are passed to Accounting)
- No assignment of records retention schedules
- No records destruction
- No inventory of records at VRC
- No longer sending hardcopy records to VRC

7.15.3 General Services / Safety

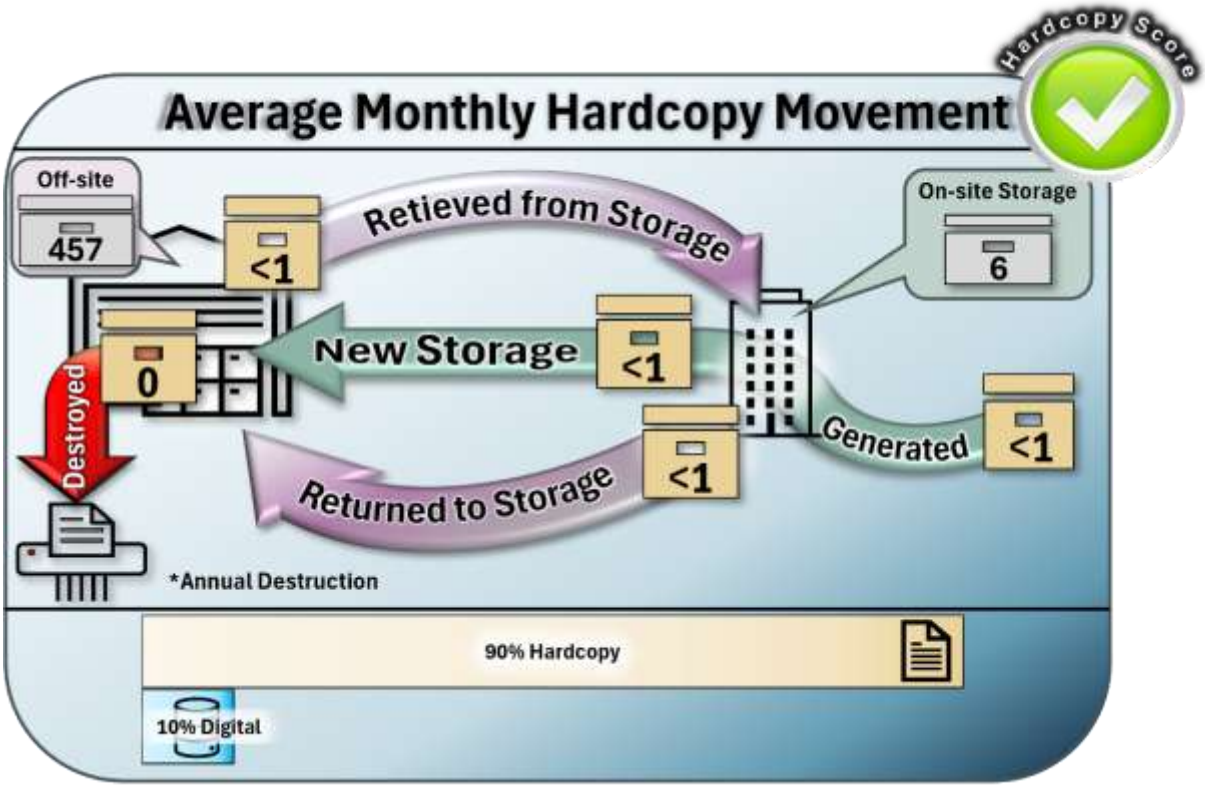
- 75% electronic / 25% hardcopy records
- No assignment of records retention schedules for electronic records
- No electronic records destruction

7.16 Health



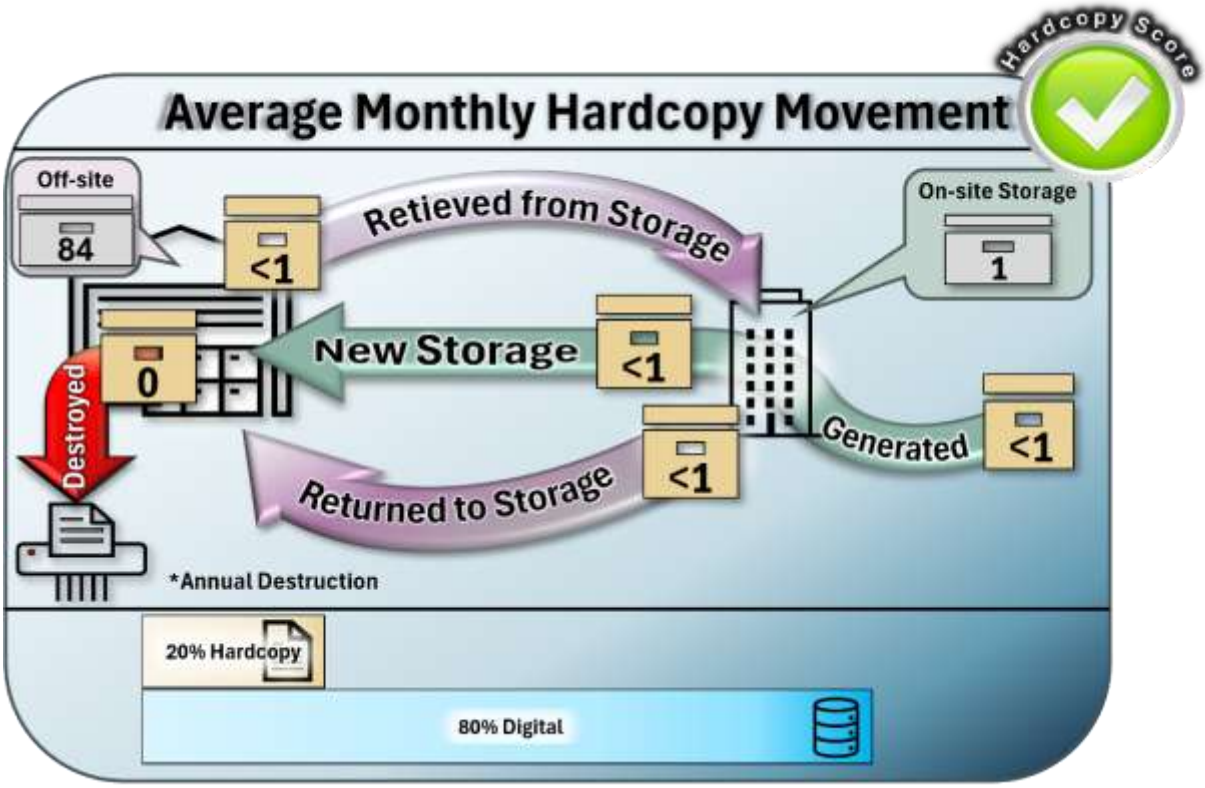
- 30% electronic / 70% hardcopy records
- The Virginia Dept of Health is planning to move to 100% digital in the next few years, including digitizing historical records
- Each Work Unit has their own Records Administrator
- No process for assigning a retention schedule or for destroying electronic records

7.17 Human Resource Management



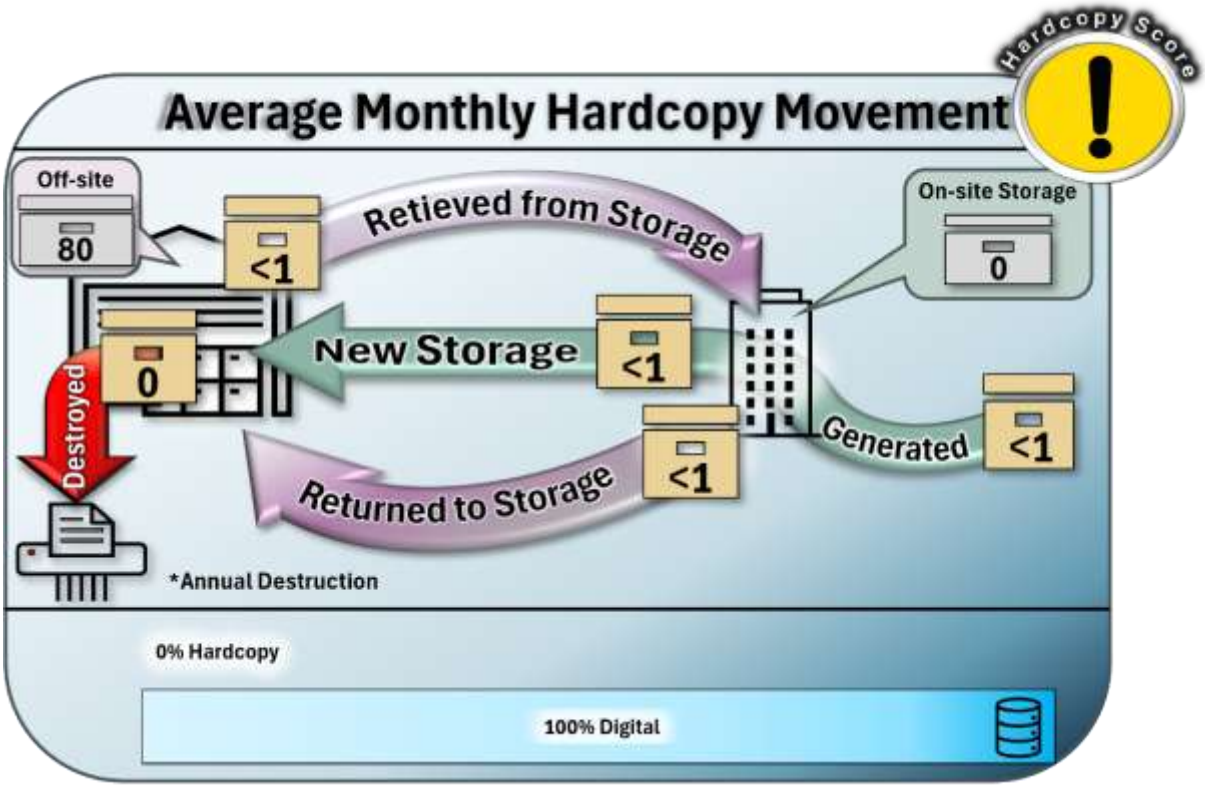
- 90% electronic / 10% hardcopy records
- No destruction of electronic records
- Spreadsheet inventory of VRC boxes
- Manual upload of documents to LaserFiche (e.g., upload annual reviews for 4600 employees, one-at-a-time)

7.18 Information Systems Technology



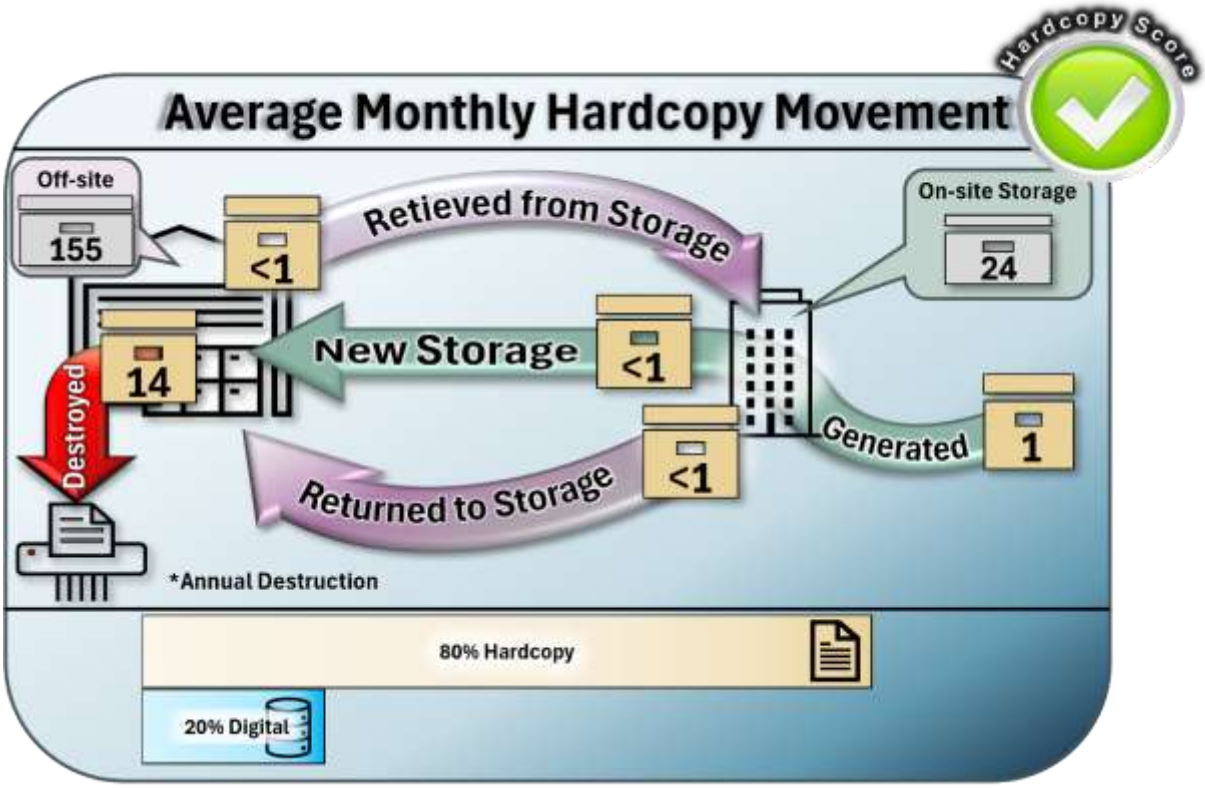
- 80% electronic / 20% hardcopy records
- No common process for assigning a retention schedule or for destroying electronic records
- VRC web portal is bad. Multiple searches return different results. Search is limited… everything gets an order record, but we cannot search by order record. On the VRC site, there are Q&A, but not everyone has access. Departments would have to send a request to get access to VRC… but this would prevent IST to see the transactions. Commonwealth Attorney can request boxes for retrieval. Description of what's in the box – we used to be able to include more text – now very limited.
- No records management system with ability to assign retention schedules and destroy expired records automatically

7.19 Internal Audit



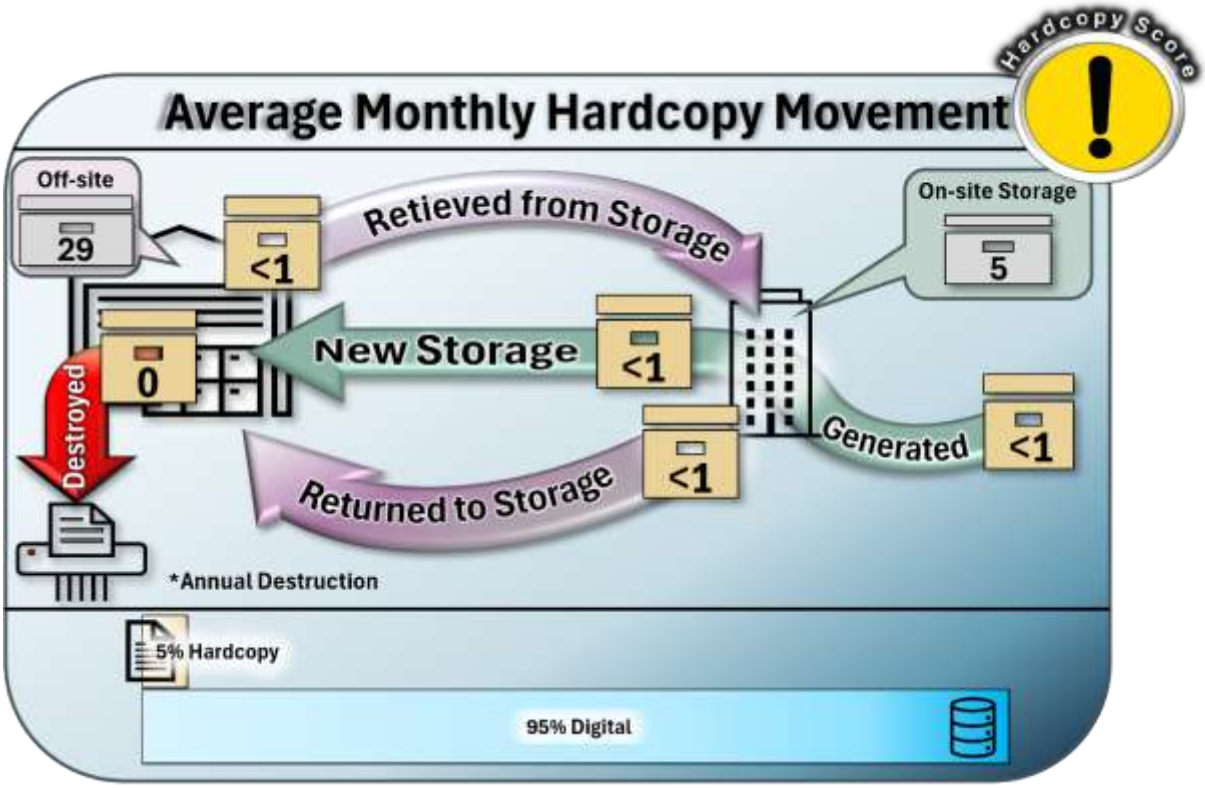
- 100% electronic records
- Standardized file naming and folder structure to store and organize electronic records, tied to retention schedules
- 1x per year they go through the files (electronic and hardcopy) and remove records that no longer need to be kept, and destroy them
- No RM-3 form for destroying locally stored records
- Boxes in VRC that they approved for destruction last year, reappear on the list for destruction this year. Were they not destroyed? Was the inventory not updated?
- Unsure of the rules (lack of policies and procedures) for destroying locally stored hardcopy and electronic records. Their experience with RM-3 forms is that Mary always initiates them.

7.20 Juvenile Justice Services



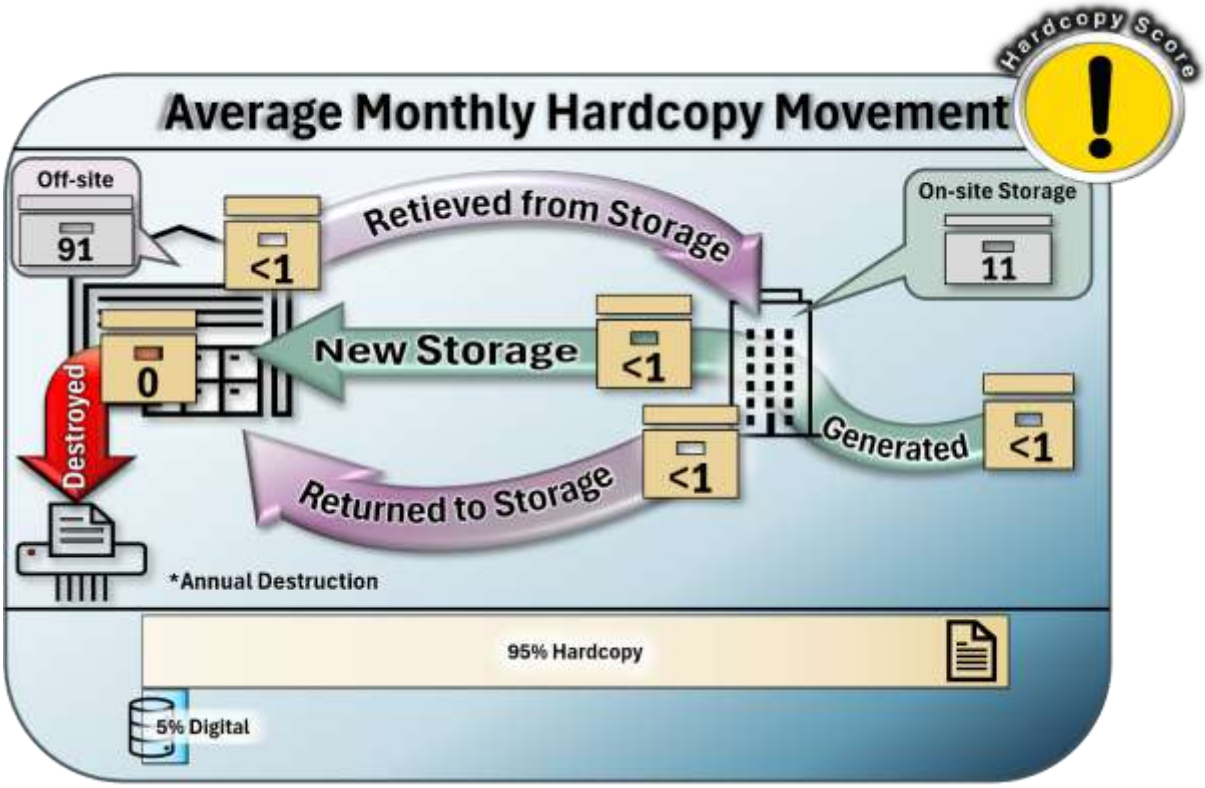
- 20% electronic / 80% hardcopy records
- They complete an RM-3 form for destroying locally stored records
- No process for assigning a retention schedule or for destroying electronic records
- They want to keep a case file together, but the file contents have different retention periods. They label the box with the longest retention period. Case file is assigned an RRS, but the contents have different RRSs
- Ownership of records can be a challenge for the Child Advocacy Center – CAC, Law Enforcement, and Child Protective Services all have interest in some records – who owns the official "record" copy?

7.21 Learning & Performance Center



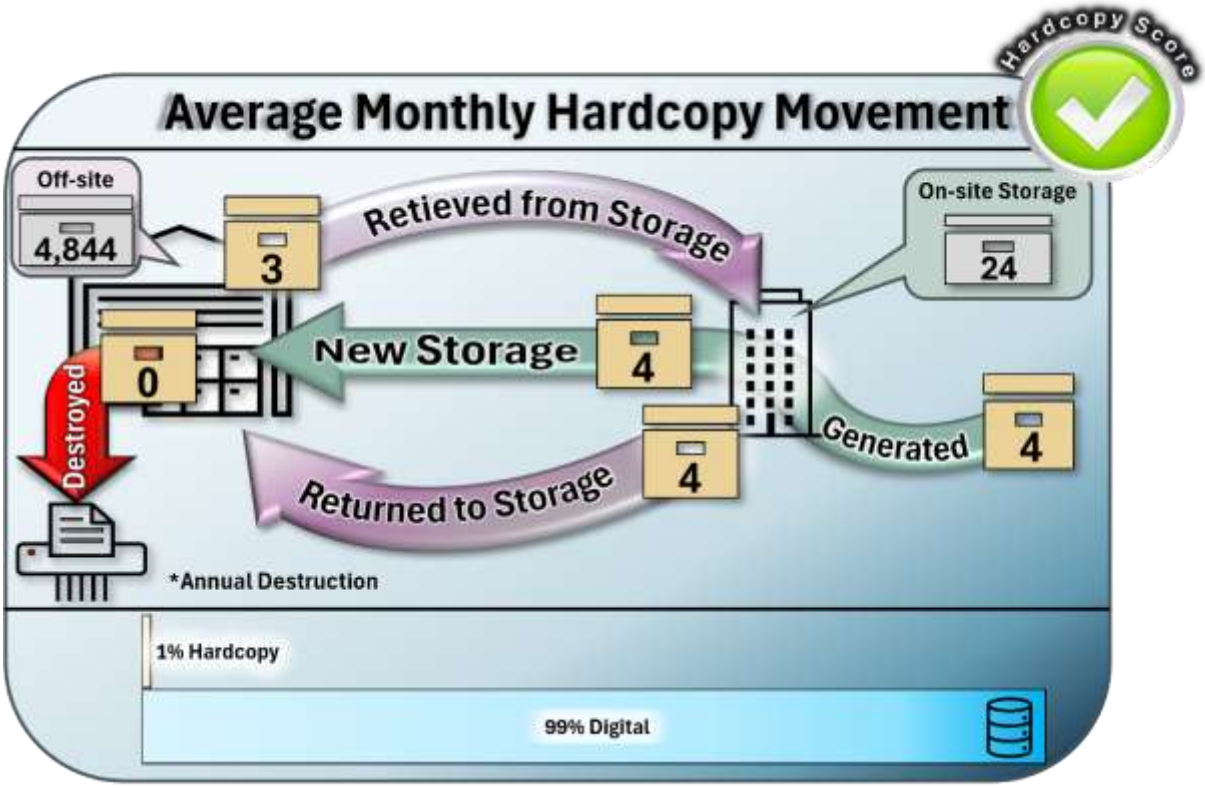
- 95% electronic / 5% hardcopy records, moving toward 100% electronic
- No off-site storage – No records stored with VRC
- Always use RM-3 form for destroying records, even for locally stored and electronic
- Investigating the best way to digitize and organize selected historical records
- Retain some records as permanent for historical purposes, which is longer that LVA schedules indicate
- The do periodic larger efforts to organize and scan older records
- They keep some records permanently, which is longer than the LVA retention schedule prescribes

7.22 Library Services



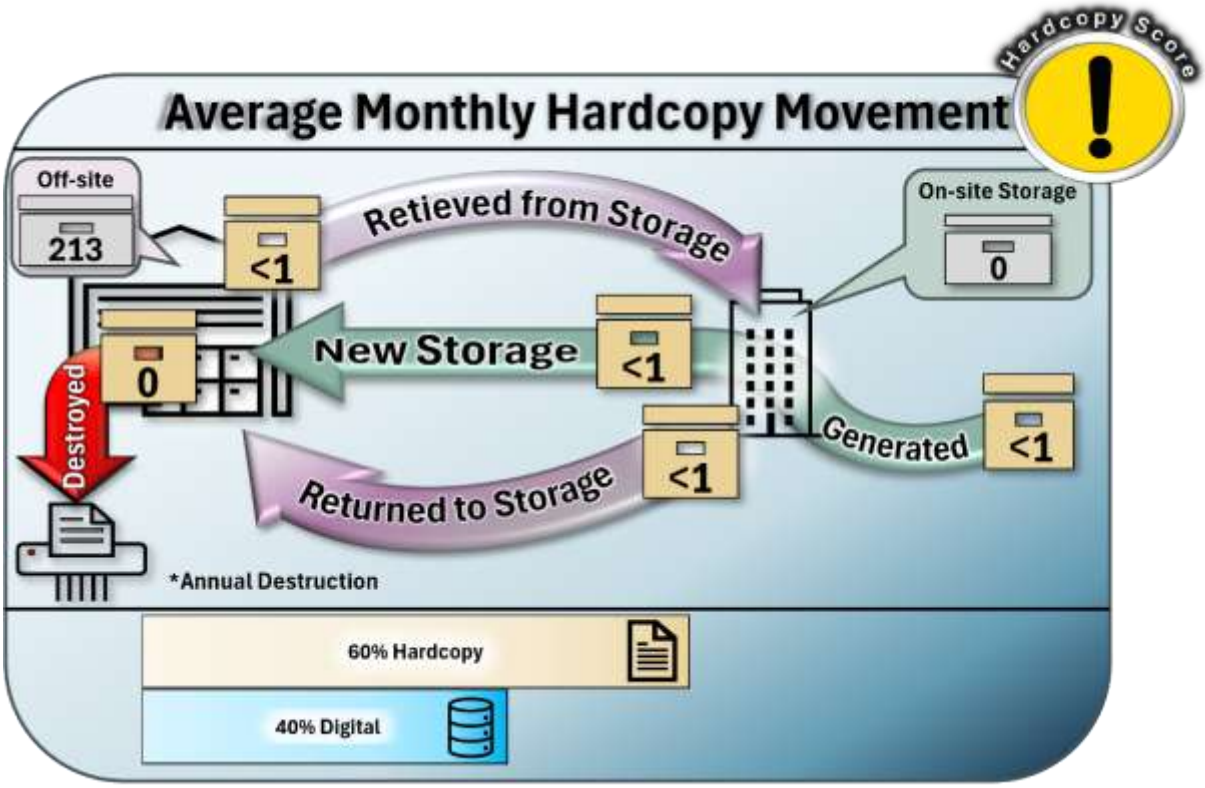
- 5% electronic / 95% hardcopy records – scans some documents and sends them to other County organizations (e.g., sends invoices to AP) for them to store as records
- 1x per year they go through the file cabinets and send to VRC
- There is some uncertainty in identifying records, hardcopy and electronic

7.23 Mental Health Support Services



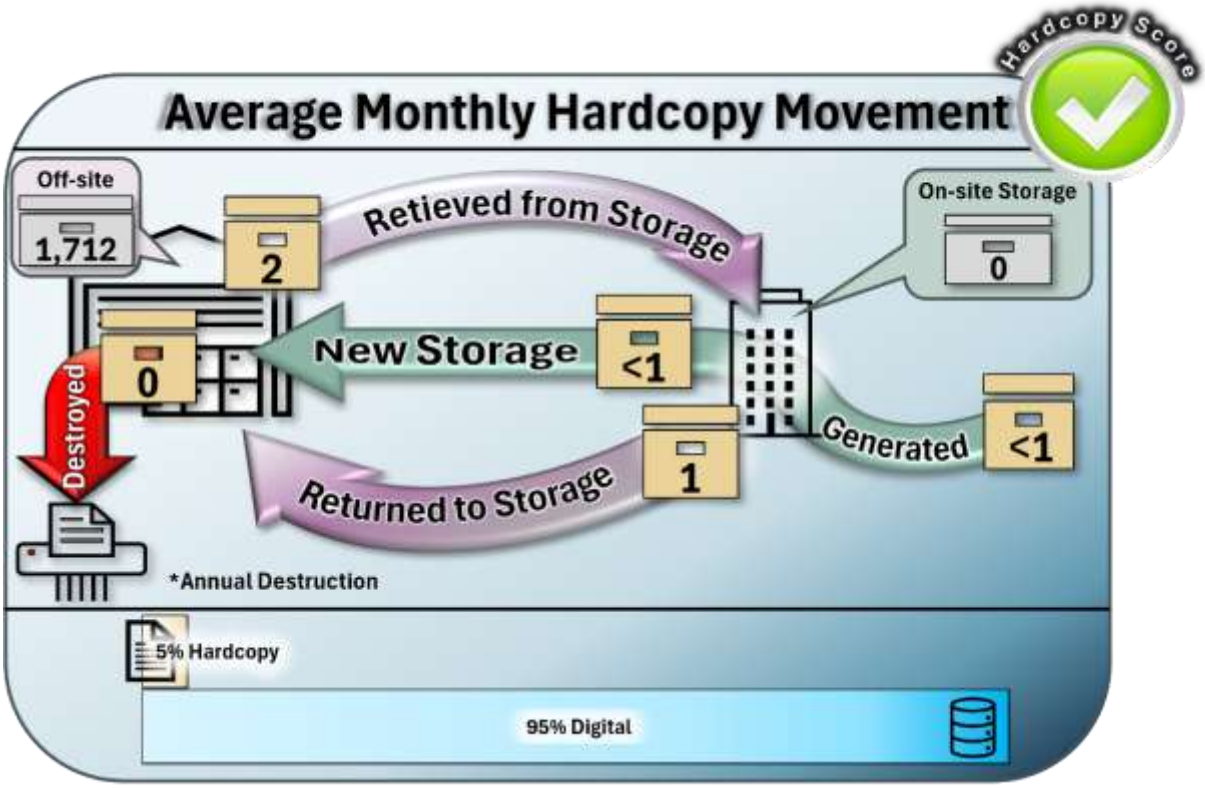
- 99% electronic / 1% hardcopy records
- No process for assigning a retention schedule or for destroying electronic records
- They are almost fully electronic, except for the Mental Health Facilities that deal primarily in hardcopy.
- Lots of ongoing digitization of newly received records
- Robust process for verifying digitized documents. Anything someone scans, someone else double checks.
- Time spent on RM = 360% of an FTE
- There are 40 different CSVs (Community Services Board – Public mental health system across the state), which periodically need to exchange records. Also, hospitals and other mental health facilities need to exchange records. There is no standardization of these records, so they are usually exchanged via hardcopy. If you could have a standard or API, you could eliminate a lot of hardcopy records.
- They use an LVA records retention schedule that requires them to retain records for 10 years after a patient is discharged, but some individuals are never discharged and some are in and out frequently. Therefore, they created their own 30-year retention period, but will require either a new record retention schedule or a formal waiver from the CIO.

7.24 Parks & Recreation



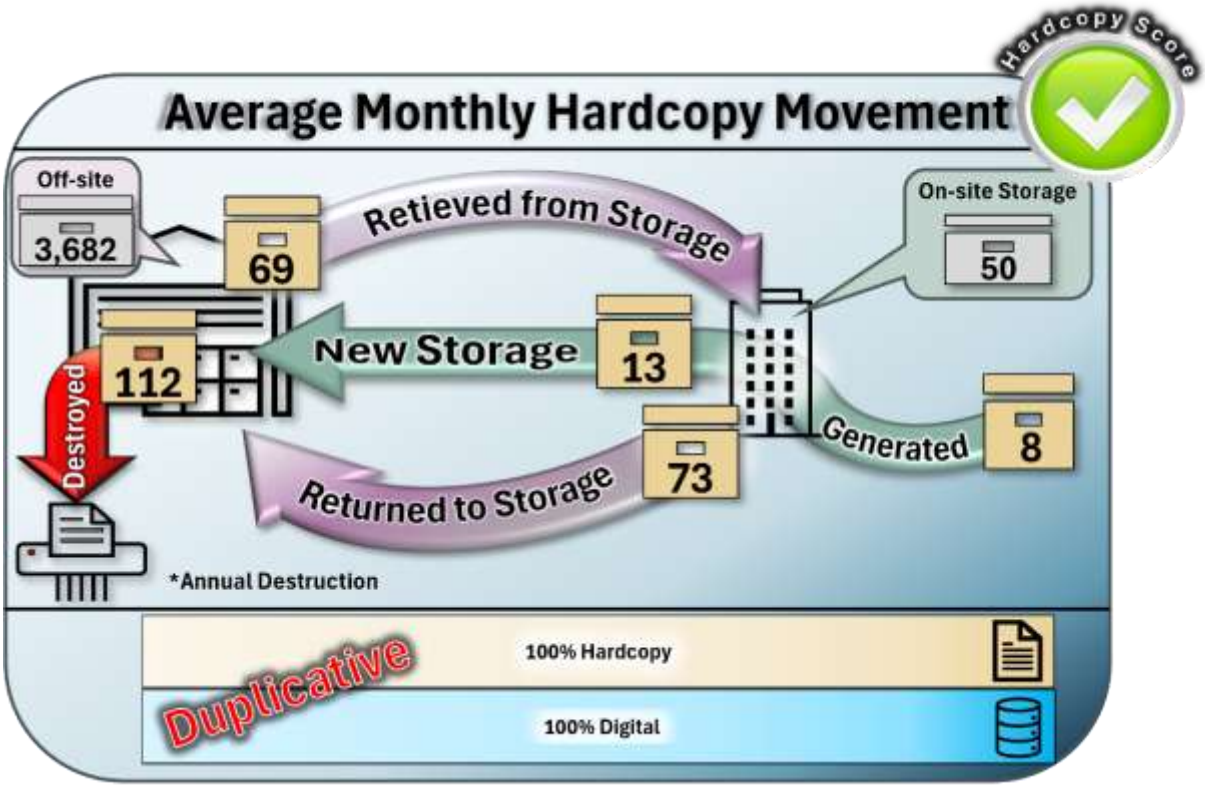
- 40% electronic / 60% hardcopy records
- 1x per year they go through the file cabinets, remove records that no longer need to be kept, and destroy them
- No process for assigning a retention schedule or for destroying electronic records
- They keep "As-Built Plans" as records. Every project has an as-built set of plans. Most are hardcopy. They cannot digitize the plans. Official record copy is with Building Inspections or Env Engineering – these departments own the record. Parks & Recreation keeps a copy for their reference. They have a room of hanging files of plans. The record copy is the permit set. There is an eCopy as well.

7.25 Planning



- 100% electronic / 10% hardcopy records – wet signatures are scanned but the hardcopy is the record
- hardcopy records at VRC are the record before ELM… ELM is the record for new stuff (2021)
- They have not started destroying records in ELM
- They can identify ELM retention schedules by doc type and file name
- They use an RM-3 form for destroying locally stored records
- Assigning a retention schedule in LaserFiche by assigning a file type and leveraging a naming convention.
- They are not destroying electronic records in LaserFiche but are working to find a way to use rules to implement deletions

7.26 Police

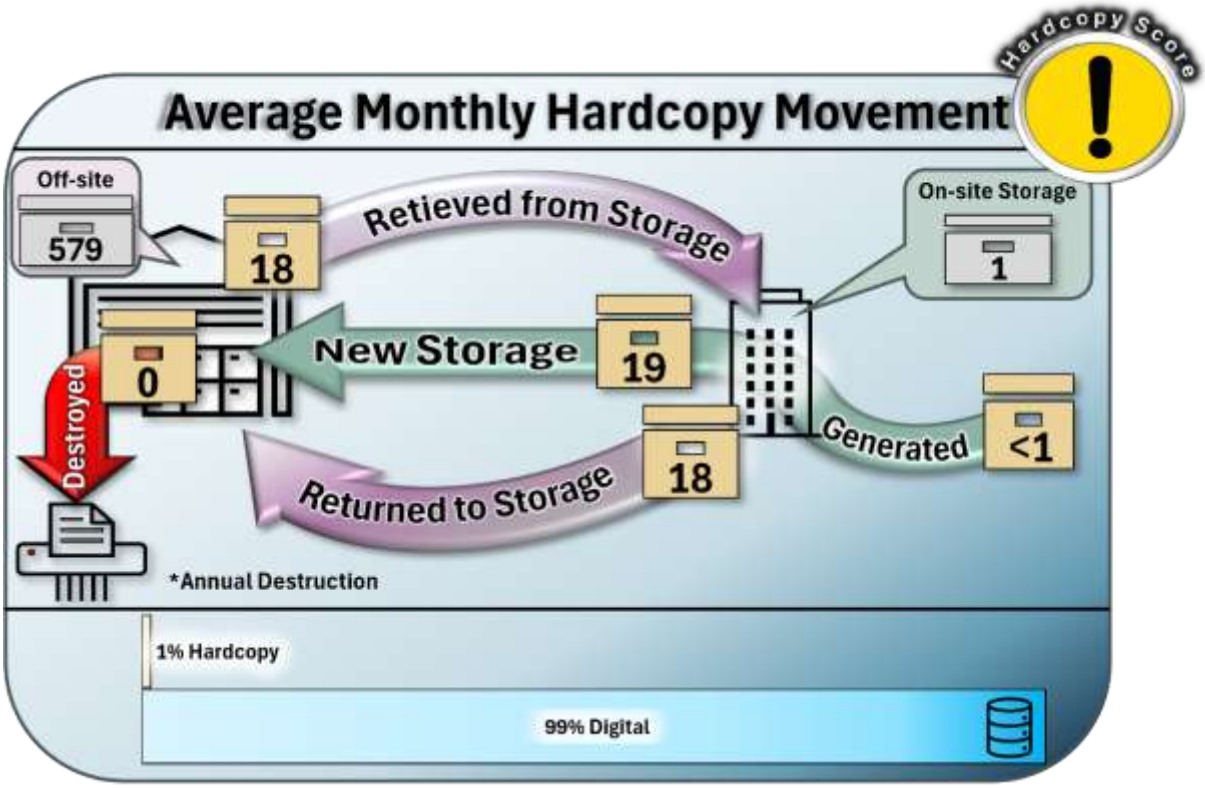


- 100% electronic / 100% hardcopy records – they keep both formats and both are considered to be records
- No process for assigning a retention schedule or for destroying electronic records
- RMS system not capable of storing all records for some cases, so cases have to be scanned twice: 1x for RMS and 1x for LaserFiche. There is no drag-and-drop between RMS and LaserFiche.
- Criminal case file records are kept permanently, which is longer than the LVA retention schedule prescribes
- Time spent on RM = 100% of one FTE

7.26.1 Police – Animal Control

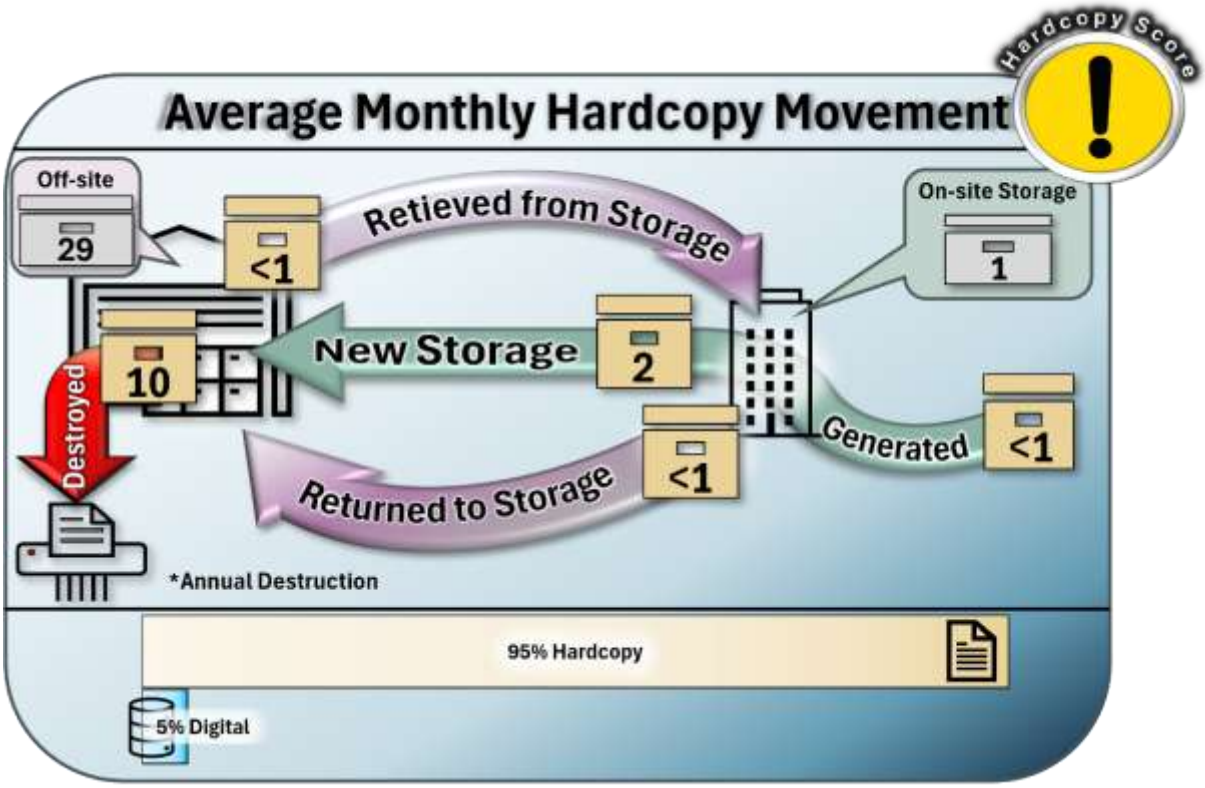
- 70% electronic / 30% hardcopy records, 20% of the hardcopy records are also stored electronically
- 1x per year they go through the file cabinets and send records to VRC
- No process for assigning a retention schedule or for destroying electronic records

7.27 Procurement



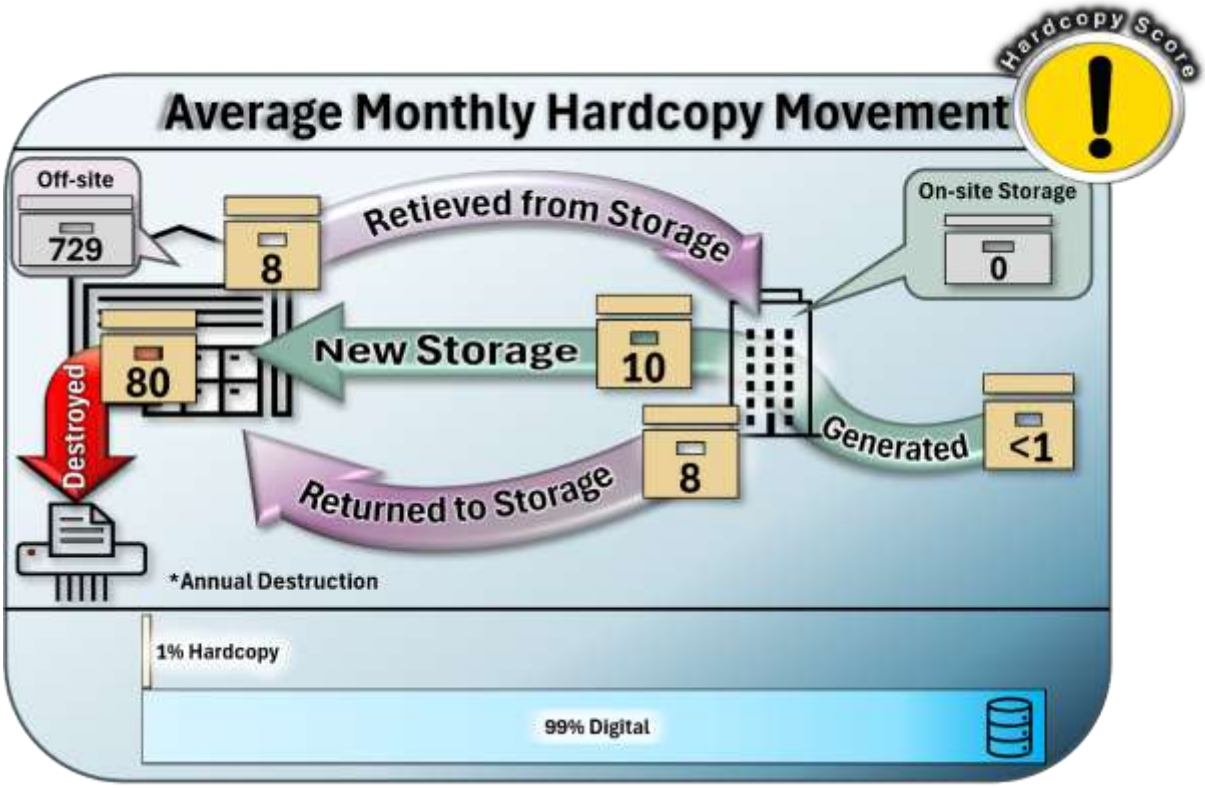
- 99% electronic / 1% hardcopy records (only keeping certain wet signatures, as directed by Legal)
- Pulled all boxes back from VRC and did major scrub this past summer
- No process for assigning a retention schedule or for destroying electronic records
- 8+ boxes are lost at VRC
- Determining when the retention period starts can be tricky – not always clear when an activity is closed / completed
- Need guidance for eRecords destruction

7.28 Real Estate Assessments



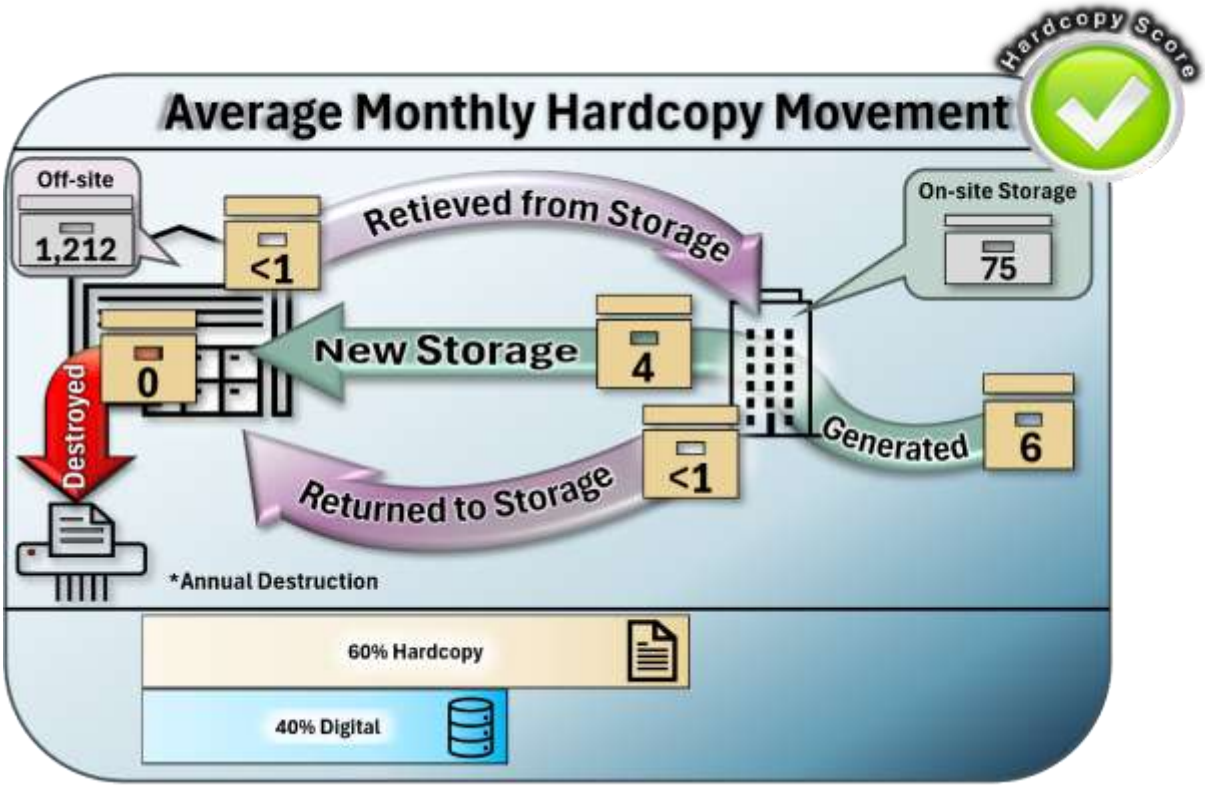
- 95% electronic / 5% hardcopy records. Some wet signature docs require hardcopy retention.
- No RM-3 form for destroying locally stored records, historically. Implementing RM-3 form for local records destruction going forward.
- No process for assigning a retention schedule or for destroying electronic records
- Digitized lots of records in preparation for recent office move
- Subject to retention schedules that don't match LVA

7.29 Risk Management



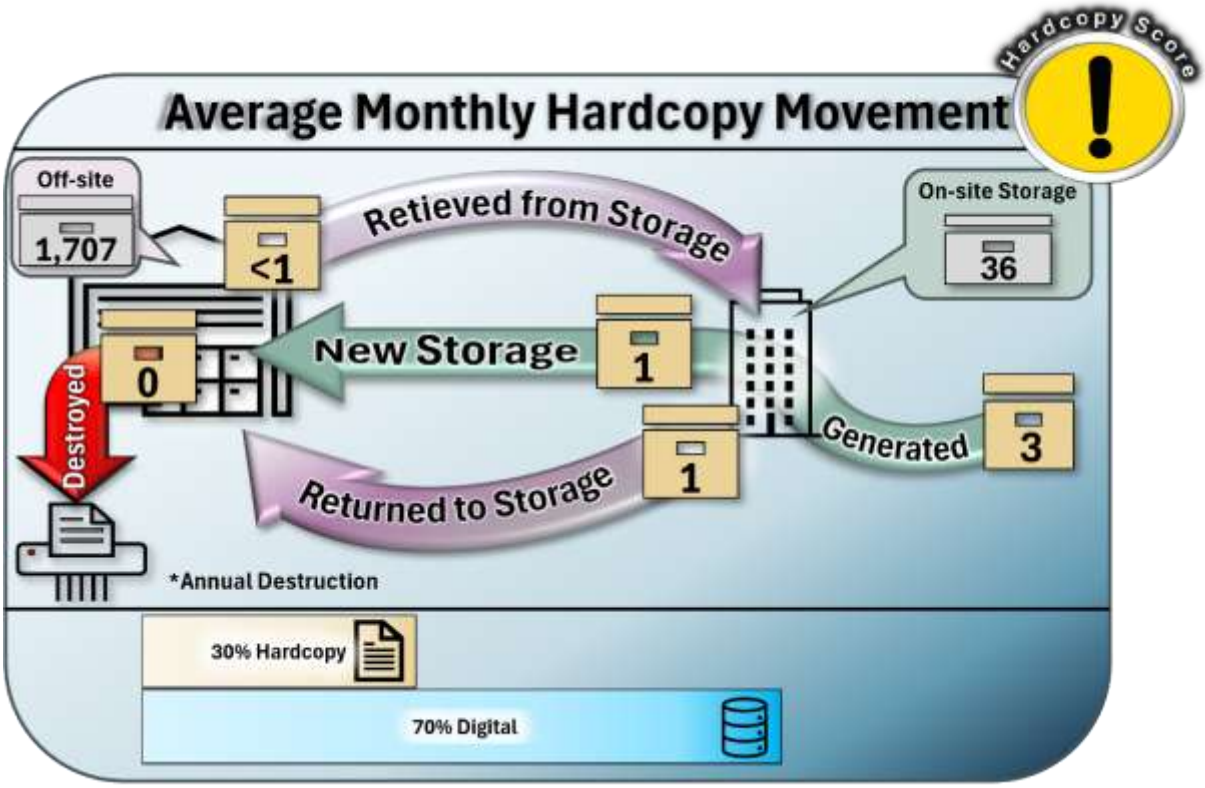
- 99% electronic / 1% hardcopy records – any hardcopies received are scanned and destroyed. A small number of 1- to 5-year retention hardcopy files are stored locally and destroyed locally, so no new VRC storage is used some old boxes remain at VRC
- No process for assigning a retention schedule or for destroying electronic records
- No RM-3 form for destroying locally stored records
- Major effort to purge old files is nearing completion
- Time spent on RM = 1-5% of one FTE for new records. Multiple FTEs involved in purging old records

7.30 Sheriff



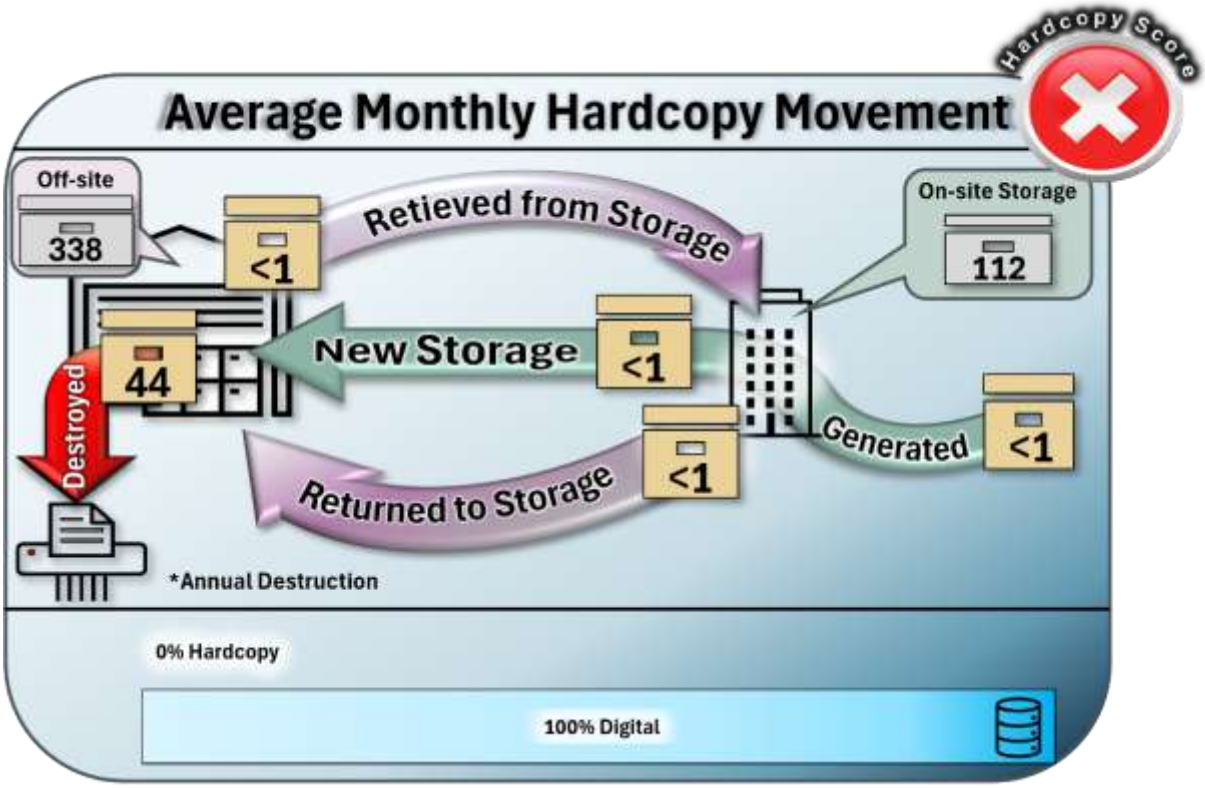
- 40% electronic / 60% hardcopy records, some records are duplicated in both formats
- Mature records management processes
- Outstanding records management in the department's IT systems

7.31 Social Services



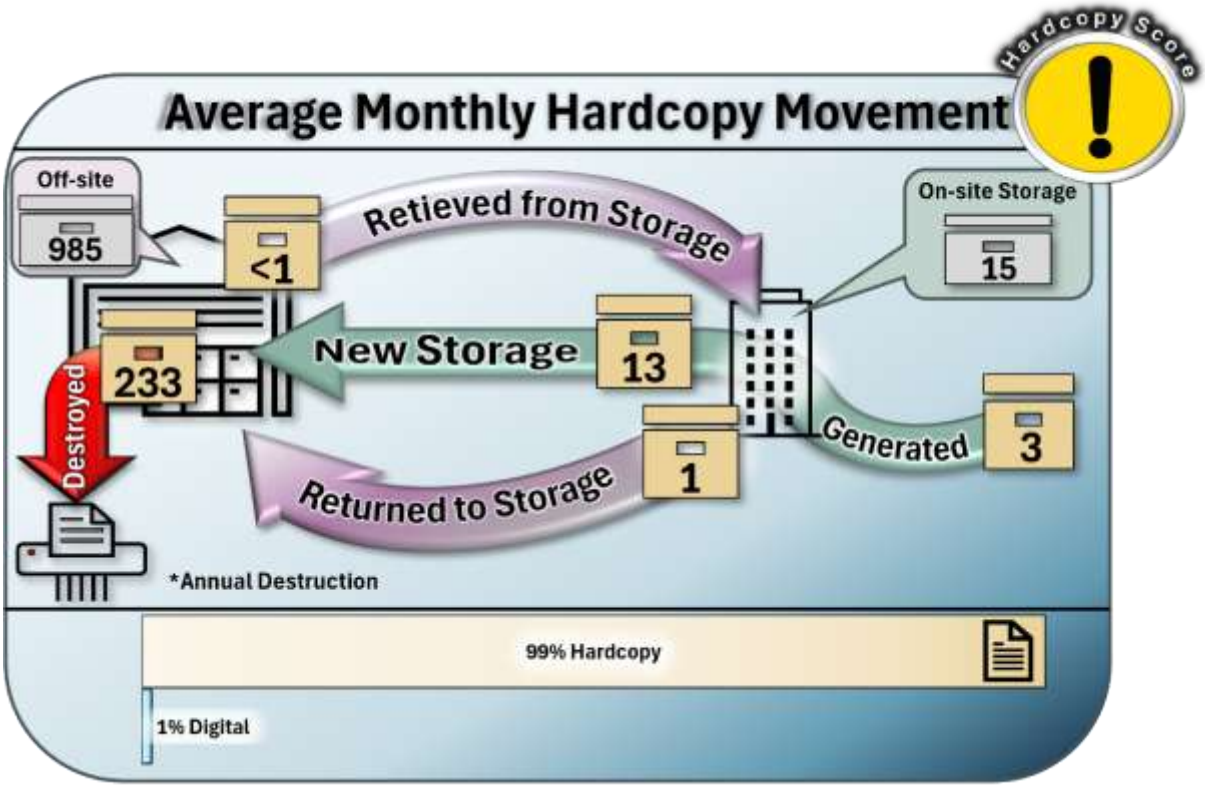
- 70% electronic / 30% hardcopy records
- Differentiating between a record and a copy of a record can be a challenge
- No process for assigning a retention schedule or for destroying electronic records

7.32 Transportation



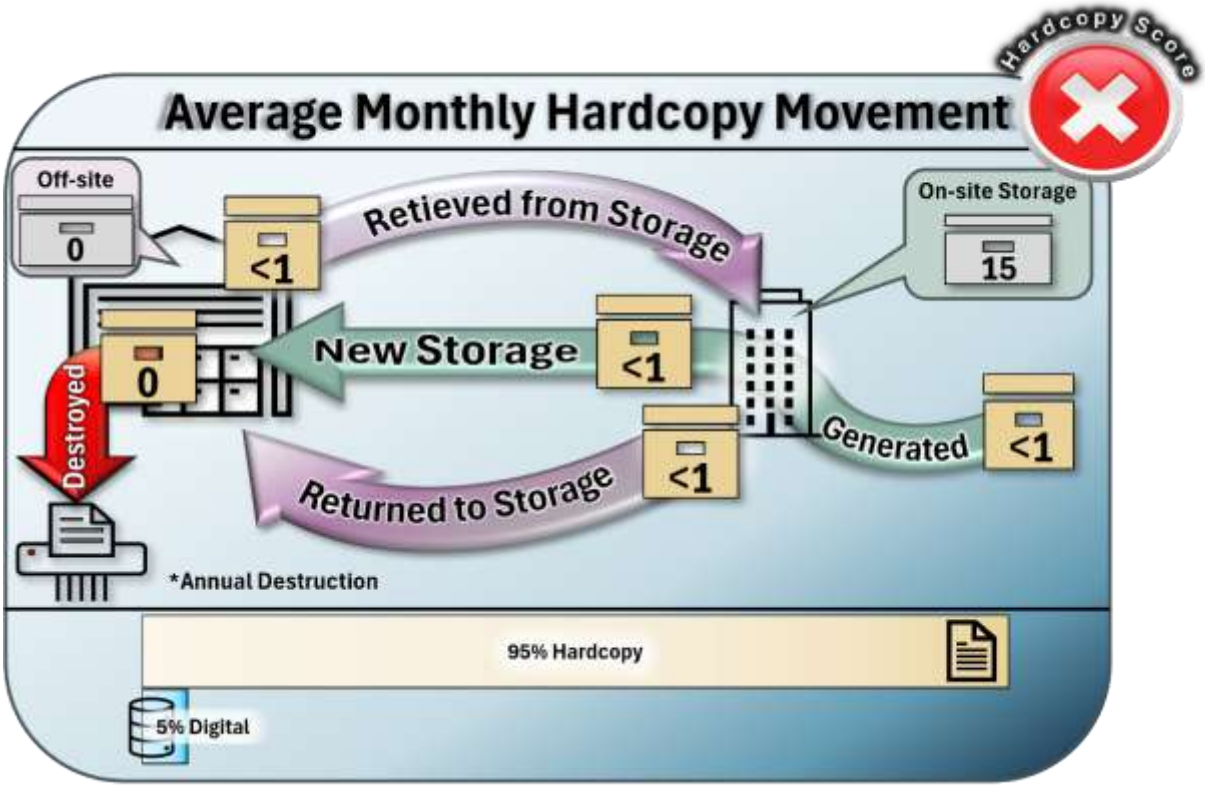
- 95% electronic / 5% hardcopy
- Not sending any boxes to VRC
- Retrieving boxes from VRC, scrubbing the contents (using RM-3 form to destroy appropriate records), and scanning the rest
- Keeping many records as permanent, but not assigning an LVA retention schedule
- No process for assigning a retention schedule or for destroying electronic records
- Email issue – Email accounts were too big, because they email files and use email to store records – Outlook is their filing system. Eventually they started having Outlook problems. Using Outlook as a filing system. They build infrastructure – multi-year projects – so they need access to email chains that document decisions. They download files, but also keep files in email.

7.33 Treasurer



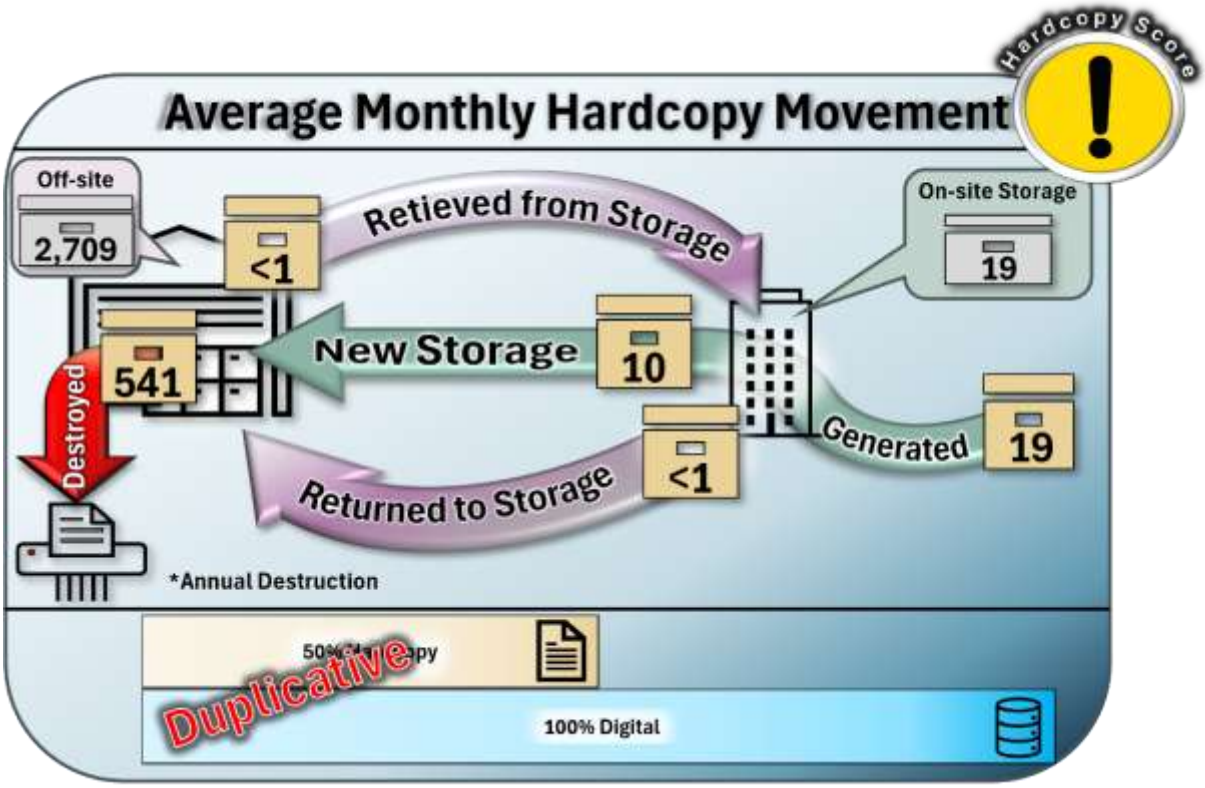
- 1% electronic / 99% hardcopy records
- No plans to increase digitization going forward
- No process for assigning a retention schedule or for destroying electronic records

7.34 Treatment Courts



- 5% electronic / 95% hardcopy records
- Recently did a scrub to eliminate boxes of records >10 years old
- Assign 10-year retention to all work products – not separating records and not assigning an LVA schedule. No plan to destroy records.
- No RM-3 form for destroying locally stored records
- No process for assigning a retention schedule or for destroying electronic records

7.35 Utilities



- 100% electronic / 50% hardcopy records
- Records that arrive in hardcopy are digitized – both copies are kept as records
- Leverage cloud backup for key files
- Moving 25-50 boxes of records per month, driven by when the records room is running out of space
- Developing a process for destroying electronic records

Appendices

Appendix A. Interview Log

The list of individuals who were interviewed to provide input to and validation of the visualization content are listed in the table, below.

Department	Record Owner / Representative	Records Liaison / POC	Interview Date
Building Inspection		Brandon Rou	8-Oct-24
Commonwealth Attorney		Ann Ward Kristie Hoskyn Frank LaRuffa	8-Oct-24
Environmental Engineering		Katherine English	9-Oct-24
Mental Health Support Services		Christine Bonner	10-Oct-24
Police	Shana Sykes	Nicole Skala	10-Oct-24
Risk Management		Kelli Payne	11-Oct-24
Fire	Graham Lathrop		11-Oct-24
Emergency Communications		Stefanie Shawkey	11-Oct-24
Fire-EMS		Millie Bishop	14-Oct-24
Community Corrections	JoAnn Carpenter	Diane Dawson	14-Oct-24
Learning & Perf Center	Janet Roberts	Tracy Connor	15-Oct-24
Human Resource Management		Mireya Guzman Liz Marker	15-Oct-24
Capital Projects	Jasna Elswick	Stefanie Creswell	15-Oct-24
Community Engagement & Resources	Kim Conley	Tracey Clock Vestine Patrick	16-Oct-24
Budget & Management		Riane Reiss	17-Oct-24
Social Services	Karen Reilly	Talara Phillips Andrew Fletcher Joni Robinson Craig Campagnplio	17-Oct-24
Internal Audit	Khara Durden		18-Oct-24
Community Enhancement	Daniel Cohen		18-Oct-24
Utilities	George Hayes	Stacey Slonaker Chris Coch	21-Oct-24
Fleet Management	John Neal	Darlean Lee	21-Oct-24
General Services (Waste Management and Recycling)	John Neal	Jennifer Oliver	22-Oct-24
Deputy Administration Office		Joy Galusha Nancy Bush	22-Oct-24
Information Systems Technology		Linda West	22-Oct-24
Transportation	Barbara Smith	Karin Wood-Bradley	23-Oct-24
Parks & Recreation		Phyllis Newcomb	24-Oct-24
Juvenile Justice Services	Marilyn Brown	Christopher Mallory	24-Oct-24
Community Enhancement		Michelle King	24-Oct-24

Department	Record Owner / Representative	Records Liaison / POC	Interview Date
Planning	Andrew Gillies	Kuzhalmozhi Sundar Steve Haasch Thomas Mendoza	25-Oct-24
Police IT		Yvonne Ferrell	25-Oct-24
Accounting	Consuela Wilson	Kimberly Collette	25-Oct-24
Police-Animal Control	Shana Sykes	Amy Estes Carrie Jones	28-Oct-24
Procurement	Stephanie Brown	Roberta Jardine	28-Oct-24
Real Estate Assessments	Melvin Bloomfield	Belinda Ashmore	28-Oct-24
Sheriff	Karl Leonard	Sara Olsen	29-Oct-24
Treasurer	Rebecca Longnaker	Jackie Atkins Kevin Payne Chartel Jefferson	29-Oct-24
Health Department	Paulette Moore		30-Oct-24
Library Services	Jennifer Stevens		1-Nov-24
IST		Mary Folkes	1-Nov-24
IST	Scott Furman		1-Nov-24
Building Inspection	Rebecca Summers		4-Nov-24
Environmental Engineering	Scott Smedley		4-Nov-24
Mental Health Support Services	Kelly Fried		5-Nov-24
Transportation	Chessa Walker		6-Nov-24
Budget & Management	Gerald Durkin		6-Nov-24
DCA for Human Services	Dr. James Worsley		6-Nov-24
General Services: Safety	Craig Willingham	Michelle Covington David Cottingham Sara Bishop	8-Nov-24
Parks & Recreation	Neil Luther		8-Nov-24
Risk Management	Trent Jones		12-Nov-24
Board of Supervisors	Susan Wilson		12-Nov-24
DCA for Community Development	Jesse Smith		14-Nov-24
Emergency Communications	Christopher Snyder		15-Nov-24
Circuit Court Clerk	Amanda Pohl		19-Nov-24
IST Infrastructure	Mike Pond		19-Nov-24
Commonwealth Attorney	Erin Barr	Kristie Hoskyn	20-Nov-24
DCA for Finance and Administration	Matt Harris		21-Nov-24
DCA for Community Operations	Clay Bowles		21-Nov-24
Treatment Courts	Geoff Easton		26-Nov-24
County Attorney		Michelle Heilman	26-Nov-24

Appendix B. Interview Questions

While interviews with Chesterfield County Records Liaisons were unique and did not follow an exact script, there were some common questions asked during each interview. These questions are listed below.

- Is Records Management training adequate?
- How much time do you spend on records management, where records management includes activities like scanning hardcopy records and sending boxes of records to Vhardcopy?
- Do you have an inventory of your records (hardcopy and electronic)?
- What is the ratio of hardcopy vs electronic records that you process every month?
- Are you subject to any audits that require you to find and retrieve records?
- Where are your hardcopy records stored (address)?
- What IT systems do you use that store electronic records? How are records identified and assigned a retention schedule in each system?
- How many boxes of hardcopy records do you produce per month?
- Do you have any department-specific records management-related policies and/or procedures?
- Which Library of Virginia (LVA) records retention schedules do you use (schedule number and record series number)?
- What are your pain points / sources of frustration when it comes to records management?
- What changes to our generic records management process, depicted in the draft visualization, are needed to make it an accurate depiction of your department-specific process?

Appendix C. Calculations for Digitizing Long-Lived Records

Option A. Digitize all off-site records

Goal: Repurpose $87K annual cost towards records management modernization (IT infrastructure, manpower)

Assumptions: *Source data: https://www.e-arc.com/scanning-imaging/*

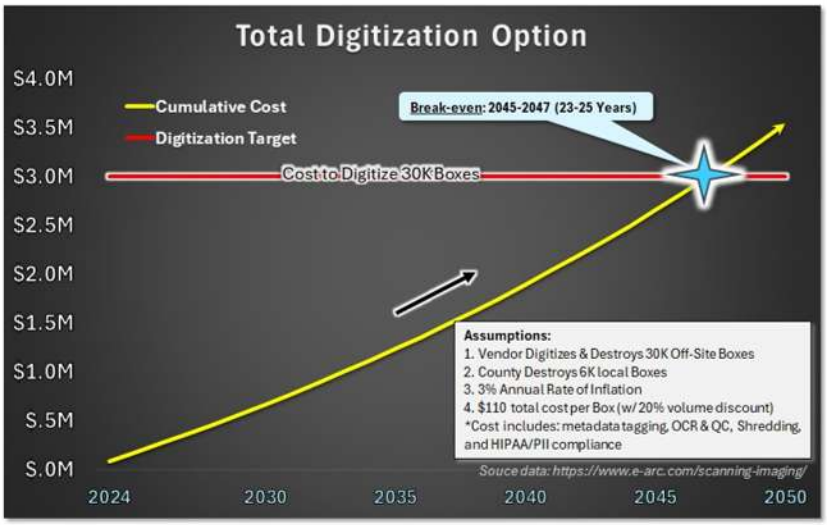
1. Vendor Digitizes & Destroys 30K Off-Site Boxes
2. County Destroys 6K local Boxes to mitigate off-site growth rate
3. 3% Annual Rate of Inflation
4. $110 total cost per Box (w/ 20% volume discount) *Includes metadata, OCR, QC, Shredding, and HIPAA/PII compliance

Upfront Cost: $3M one-time cost to digitize 30K hardcopy boxes

Break-Even Point: 23-25 years using the above assumptions

Digitization Estimate (Everything by Private Vendor)

Year	Annual Cost	Cumulative Cost	Digitization Target
2024	$ 87,000	$ 87,000	$ 3,000,000
	$ 89,610	$ 176,610	$ 3,000,000
	$ 92,298	$ 268,908	$ 3,000,000
	$ 95,067	$ 363,976	$ 3,000,000
	$ 97,919	$ 461,895	$ 3,000,000
	$ 100,857	$ 562,752	$ 3,000,000
2030	$ 103,883	$ 666,634	$ 3,000,000
	$ 106,999	$ 773,633	$ 3,000,000
	$ 110,209	$ 883,842	$ 3,000,000
	$ 113,515	$ 997,358	$ 3,000,000
	$ 116,921	$ 1,114,278	$ 3,000,000
2035	$ 120,428	$ 1,234,707	$ 3,000,000
	$ 124,041	$ 1,358,748	$ 3,000,000
	$ 127,762	$ 1,486,510	$ 3,000,000
	$ 131,595	$ 1,618,106	$ 3,000,000
	$ 135,543	$ 1,753,649	$ 3,000,000
2040	$ 139,609	$ 1,893,258	$ 3,000,000
	$ 143,798	$ 2,037,056	$ 3,000,000
	$ 148,112	$ 2,185,168	$ 3,000,000
	$ 152,555	$ 2,337,723	$ 3,000,000
	$ 157,132	$ 2,494,854	$ 3,000,000
2045	$ 161,846	$ 2,656,700	$ 3,000,000
	$ 166,701	$ 2,823,401	$ 3,000,000
	$ 171,702	$ 2,995,103	$ 3,000,000
	$ 176,853	$ 3,171,956	$ 3,000,000
	$ 182,159	$ 3,354,115	$ 3,000,000
2050	$ 187,623	$ 3,541,738	$ 3,000,000



Option B. Digitize the most costly off-site records

Goal: Digitize the off-site records of the top twelve costliest departments (9-10K boxes)

Assumptions: *Source data: https://dashboard.diglib.org/*

1. Hire three interns (minimum wage of $12.41/hour)
2. Departments reduce off-site storage by 20%
3. Spot checks < 10% of documents for scanning errors
4. < 50% of documents require preparation (fastener removal)
5. Basic quality control (reviews < 20% of digital files for accuracy)

6. Minimal post-processing (de-skewing, etc.)
7. Automated descriptive and structural metadata creation

Upfront Cost: Budget for three interns working 250 hours / year for 16 years (about $8-9K/Year)

Break-Even Point: 16 years

Digitization Estimate (1x Box)

Source: https://dashboard.diglib.org/

1x 1 cu-ft document box
Intern (Minimum Wage) $12.41/Hour
Number of scans: 1,200 documents
Total Estimated Time: 7.91 hours
Total Estimated Cost: $86.94

Cost to digitize one box per day

Task	Sub-Task	% Materials	Time (minutes)	Cost
Image Capture	Sheet feeding scanner		86.64	$18.19
Preparation of Materials	Condition Review	10%	14.6	$3.06
Preparation of Materials	Fastener Removal	50%	262.18	$55.06
Post Processing	Alignment	10%	35.47	$7.45
Post Processing	Clean Up	0%	0	$0.00
Quality Control	Level 1	20%	75.51	$3.17
		Total:	474.4	$86.94

Cost to Store a Digital Record

Source: https://azure.microsoft.com/en-ca/pricing/details/storage/f/

Assumptions:
1x Hardcopy Record digitized to an 8 MB PDF
1024 MB = 1 GB
500 TB/Month (Hot, LRS Storage): $0.01666 per GB
Cost of one PDF Record: $0.00013
Cost of an equivilent "Box" of 1200 eRecords: $0.156

Hot: An online tier optimized for storing data that is accessed or modified frequently. The hot tier has the highest storage costs, but the lowest access costs.

LRS: Locally redundant storage

			3x Interns working 250 days / Yr	
Department	Annual Cost	% of Total Cost	Boxes Digitized	Years to Digitize
Police	$ 4,215	25%	2,946	4
Commonwealth Attorney	$ 2,838	17%	1,410	3
Procurement	$ 1,186	7%	463	0.6
Mental Health	$ 1,176	7%	3,875	5
Risk Management	$ 713	4%	583	1
Utilities	$ 647	4%	2,167	3
County Attorney	$ 612	4%	778	1
Total:	$ 11,386	67%	12,222	16 Years

Year	Cummulative Savings	Cummulative Cost	Net ROI
1	$ 1,300	$ 9,300	$ (8,000)
2	$ 4,200	$ 18,600	$ (14,400)
3	$ 8,100	$ 27,900	$ (19,800)
4	$ 12,300	$ 37,200	$ (24,900)
5	$ 18,300	$ 46,500	$ (28,200)
6	$ 25,700	$ 55,800	$ (30,100)
7	$ 34,600	$ 65,100	$ (30,500)
8	$ 44,200	$ 74,400	$ (30,200)
9	$ 54,500	$ 83,700	$ (29,200)
10	$ 65,300	$ 93,000	$ (27,700)
11	$ 76,500	$ 102,300	$ (25,800)
12	$ 88,900	$ 111,600	$ (22,700)
13	$ 101,700	$ 120,900	$ (19,200)
14	$ 115,100	$ 130,200	$ (15,100)
15	$ 129,100	$ 139,500	$ (10,400)
16	$ 144,400	$ 148,800	$ (4,400)
	$ 161,000	$ 158,100	$ 2,900

Pay three interns to digitize 12K boxes of the top seven costliest departments, sequentially, over the next 16 years to save $160K+ ($11K+ per year)

Appendix D. Record Format Considerations

FILE FORMATS COMPARISON

This website by Cloudinary provides an excellent overview of the most common electronic record formats, along with descriptions, pros, and cons.

The section, below, shows the policies / direction related to electronic record file formats from several highly respected organizations.

HIGHLY RESPECTED ORGANIZATIONS' APPROACHES TO FILE FORMATS

Code of Federal Regulations §1236.48 File Format Requirements (Excerpt)

(a) The agency must encode, retain, and transfer digitized records in one of the following file formats, either uncompressed or using one of the specified compression codecs in tables 1 and 2 to this section.

> *(1) Agencies that combine multiple uncompressed TIFF images into PDF/A files using JPEG2000 compression must perform the quality inspection step specified in 1236.46(d) against the resulting PDF/A files.*
>
> *(2) When using JPEG 2000 visually lossless compression, agencies must determine the amount of compression to apply, not to exceed 20:1, by performing tests and visually evaluating for compression artifacts that obscure or alter the information content.*

(b) The agency must encode, retain, and transfer digitized permanent paper records in one of the following file formats, either uncompressed or with one of the compression codecs specified in table 1 to this paragraph (b).

Table 1 to Paragraph (b)—File Format Requirements for Digitized Permanent Paper Records Table

Format name and version	***Acceptable compression codecs***
Tagged Image File Format 6.0 (TIFF)	*Uncompressed, Deflate (ZIP).*
JPEG2000 part 1 (ISO/IEC 15444-1:2019)	*JPEG 2000 part 1 core coding system lossless compression. Agencies may use up to 20:1 visually lossless compression.*
Portable network graphics 1.2 (PNG)	*Deflate (ZIP).*

Table 1 to Paragraph (b)—File Format Requirements for Digitized Permanent Paper Records Table	
Format name and version	***Acceptable compression codecs***
Portable Document Format (PDF) PDF/A (Select any open source version of PDF/A that meets project requirements). However, do not use the attachments feature in PDF/A-3 or PDF/A-4	*Deflate (ZIP), JPEG 2000 part 1 core coding system lossless compression. Agencies may use up to 20:1 visually lossless compression.*

(c) The agency must encode, retain, and transfer digitized photographic print records in one of the following file formats, either uncompressed or with one of the compression codecs specified in the table 2 to this paragraph (c).

(1) For a series of predominantly textual records with interspersed photographic prints, use the formats in table 1 to paragraph (b) of this section for paper records. All photographic prints must be digitized according to the standards in § 1236.50.

(2) For a series of predominantly printed photographs, including those with paper records interspersed, use the file formats in table 2 to this paragraph (c) for photographic print records.

(3) However, the agency must not transcode, or interpolate (upsample) files anywhere in the workflow.

Table 2 to Paragraph (c)—File Format Requirements for Digitized Permanent Photographic Print Records Table	
Format name and version	***Acceptable compression codecs***
TIFF 6.0	*Uncompressed, Deflate (ZIP).*
JPEG2000 part 1 (ISO/IEC 15444-1:2019)	*JPEG 2000 part 1 core coding system lossless compression. Agencies may use up to 20:1 visually lossless compression.*
Portable network graphics 1.2 (PNG)	*Deflate (ZIP).*

Library of Congress

- Preferred electronic formats for text-based records: PDF/UA, PDF/A, PDF/X

- Preferred electronic formats for photographs: TIFF, JPEG2000, PNG, JPEG/JFIF
- Preferred electronic formats for graphic images: TIFF, JPEG2000, PNG, JPEG/JFIF

National Archives and Records Administration (NARA)

All federal agencies are subject to NARA records management policies.

- Preferred record formats for different types of records
- Acceptable Formats for Digitized Permanent Paper Records: TIFF, JPEG2000, PNG, PDF/A
- Acceptable Formats for Digitized Permanent Photographic Print Records: TIFF, JPEG2000, PNG

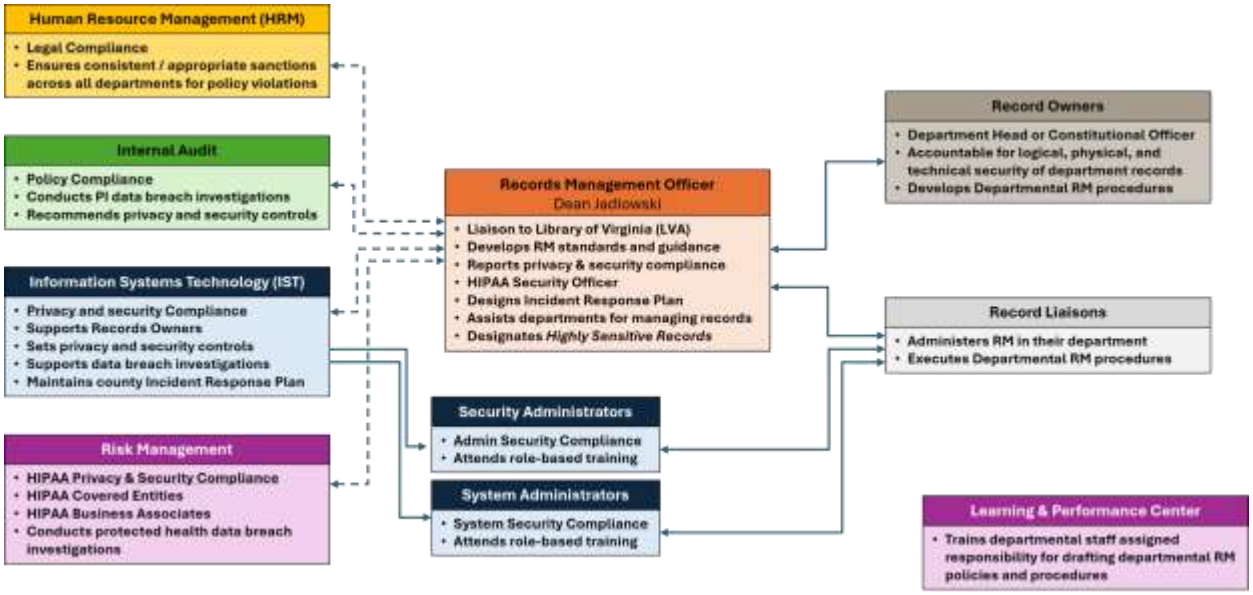
Smithsonian

- Text/word processing applications
  - Preferred Format(s): PDF/A, PDF
  - Secondary Format(s): RTF (text), TXT, XML with schema
- Images
  - Preferred Format(s): TIFF
  - Secondary Format(s): JPG, DNG, PNG, JP2
- Graphics
  - Preferred Format(s): TIFF (uncompressed)
  - Secondary Format(s): PDF

Appendix E. Mapping Key Reads & Observations to Recommendations

		Key Reads & Observations																
		Records Identification		Records Storage & Retrieval					Records Retention & Destruction						Records Digitization			
Grouping	Recommendations	Maintaining Multiple Copies of Records	Lack of Separation for Records vs. Non-Records	Records Retained Beyond Retention Period	Successful Digitization & Destruction	Hardcopy Retention Required for Signed Records	Confusing Records Transfer Process	Lost Records	Inability to Confirm Correct Records Destruction	Lack of Retention Schedule Assignment	Lack of eRecord Retention Period Tracking	Lack of eRecord Destruction	No RM-3 for Local Records Destruction	Lack of Authority & Resources	Lack of Records Management in Software Platforms	eRecord File Format Confusion	Police RMS-LaserFiche Records Duplication	Digitization Prioritization
Enhance Records Management Oversight & Leadership	Enhance Records Management Officer Roles & Responsibilities	●		●		●				●	●	●	●	●	●	●		●
	Provide Training	●		●		●	●			●	●	●	●					
	Conduct Audits and Provide Support	●		●		●	●			●	●	●	●		●			
	Control Off-Site Storage Costs	●		●	●	●		●								●	●	●
	Define and Track Measures & Metrics			●						●	●	●						
	Develop Enhanced Records Transfer Process & Instructions						●			●	●	●	●					
	Conduct Awareness Campaign	●		●		●				●	●	●	●					
	Relocate Records Management Officer Authority													●				
Reduce the Quantity of Hardcopy Records	Investigate Perceived Need for Duplicate Records	●				●												
	Review, Digitize, and Destroy Hardcopy Records	●		●	●	●				●							●	●
	Investigate Wet Signature Record Requirements	●			●	●												
	Define Electronic Record Format Requirements															●		
	Exchange More Electronic Records, Fewer Hardcopy				●													
Comply with LVA Records Retention Requirements	Develop Records Locator Inventory (LVA RM-2D Form)		●							●								
	Manage LVA Records Retention Schedule Deviations			●														
	Add Box# to LVA Records Destruction Workflow								●									
Improve Electronic Records Management	Enhance County-wide Electronic Records Management Policy	●	●	●						●	●	●			●	●		●
	Define Records Management Plans for Software Platforms									●	●	●			●			
	Pursue Automated Retention Schedule Assignment and Records Destruction		●	●						●	●	●			●			

Appendix F. Records Management Roles & Responsibilities



Example Records Management Roles and responsibilities derived from Library of Virginia policy template:

Chesterfield County Administrator (or designee)

- Designate a records officer
- Develop policies and procedures for records management
- Conduct a survey or inventory of records in all formats
- Establish and follow Library of Virginia–approved records retention and disposition schedules
- Use technology, as appropriate, to create, store, and retrieve records
- Destroy records at the end of their retention period and document that destruction
- Store inactive records in a cost-effective and secure location
- Identify and protect essential records
- Identify and preserve permanent records
- Develop a records emergency response plan to protect and recover all records
- Implement filing and indexing systems

Records Officer

- The designated records officer is responsible for the development, implementation, and ongoing coordination of the records management program to meet regulatory requirements

- Develop and implement a records management program based on Library of Virginia best practices
- Provide training in records management procedures, practices, and use of appropriate forms, as necessary. As needed, collaborate with departmental Records Coordinators to help maintain an effective records management program
- Distribute Library of Virginia-approved retention schedules information to appropriate staff members. Assist staff in surveying records that are unique to their respective departments
- Assist in identifying records ready for destruction, approve the accurate and timely destruction of records
- Identify all essential and archival records
- Participate in decisions concerning records reformatting and/or storage
- Coordinate and approve the transfer of permanent, archival records
- Coordinate the development of a plan to help ensure the protection / recovery of records in the event of a disaster or any other unplanned incident
- Receive periodic updates from the Library of Virginia on important records management issues within the Commonwealth via the Virginia Records Officer Listserv (VA-ROL)

Records Owner

- The heads of departments, divisions, and/or sections are responsible for ensuring that procedures and programs within their areas of responsibility meet the requirements of the county's records management program relative to record identification, generation, control, maintenance, processing, storage, and disposition

Records Liaison

- Records Liaisons are responsible for assisting in the design, implementation, and management of the county records management program by serving as liaisons between their respective departments and the county's designated records officer
- Being familiar with the county's internal records management policy
- Developing departmental records management procedures and practices
- Educating staff within the organizational unit in understanding sound record management practices
- Restrict access to confidential records and information.
- Coordinate the destruction of records with the records officer

Appendix G. Records Management Reference Documents

Virginia Public Records Act
Code of Virginia, Title 42.1 Libraries, Chapter 7 (§§ 42.1-76 thru 42.1-91)

Virginia Public Records Act, A Guide
https://www.lva.virginia.gov/agencies/records/documents/vpra.pdf

Virginia Public Records Management Manual, October 2024
https://www.lva.virginia.gov/agencies/records/manuals/vprmm.pdf

Library of Virginia Retention Schedules
https://www.lva.virginia.gov/agencies/records/retention.asp

Creating Policy and Procedures
https://www.lva.virginia.gov/agencies/records/policy.htm

Electronic Records
https://www.lva.virginia.gov/agencies/records/electronic/index.htm

Tip Sheets
https://www.lva.virginia.gov/agencies/records/tips/

Timely Destruction of Records
https://www.lva.virginia.gov/agencies/records/timely.asp

Virginia Records Officer Listserv (VA-ROL)
https://www.lva.virginia.gov/agencies/records/varol.htm
An electronic mailing list for discussion of records management topics by the Records Officers of Virginia's local and state governments. Through VA-ROL, participants can share problems and the solutions they've found in managing the records of their organization, get notification of updated General Schedules as soon as revised, and receive current workshop schedules or other timely information.

LVA Records Management Analyst Contacts
https://www.lva.virginia.gov/agencies/records/analyst-contacts.htm

Chesterfield County Administrative Policies and Procedures 7-12, Records Management, April 4, 2023

Chesterfield County Administrative Policies and Procedures 7-15, Records and Information Management, June 30, 2016